

A Project Report On

Placement Preparation Portal

Submitted in partial fulfillment of the requirement for the
award of the degree

Master of Computer Applications
(MCA)

Academic Year 2026 – 27

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Marwadi
University
Marwadi Chandarana Group



Faculty of Computer Applications (FoCA)

Certificate

This is to certify that the project work entitled
Placement Preparation Portal
submitted in partial fulfillment of the requirement for
the award of the degree of
Master of Computer Applications (MCA)
of the

Marwadi University

is a result of the bonafide work carried out by

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Dr. Rijwan Khan
Dean

DECLARATION

I hereby declare that the project entitled “**Placement Preparation Portal**” has been developed by me as part of the academic requirements for the **Master of Computer Applications (MCA)** programme at **Marwadi University**.

I further declare that the work presented in this project report is based on my own efforts, study, design, and development. The project has been developed using technologies including **React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, JWT, bcryptjs, and Axios**.

I have made every effort to ensure that the information, analysis, design, and implementation details presented in this report are accurate to the best of my knowledge. This project report has not been submitted previously to any other university or institution for the award of any degree or qualification.

Place:

Date:

Hardik Jethava
Hardik Vala

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92500584026 Signature: _____



ACKNOWLEDGEMENT

It is indeed a great pleasure to express our thanks and gratitude to all those who helped us. No serious and lasting achievement or success one can ever achieve without the help of friendly guidance and co-operation of so many people involved in the work.

We are very thankful to our guide **Dr.Divyakant Meva**, the person who makes us to follow the right steps during our project work. We express our deep sense of gratitude to for his /her guidance, suggestions and expertise at every stage. A part from that his/her valuable and expertise suggestion during documentation of our report indeed help us a lot.

Thanks to our friend and colleague who have been a source of inspiration and motivation that helped to us during our project work.

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1. Introduction to Project Definition

The Placement Preparation Portal is a web-based platform designed to provide students with a complete and structured environment for preparing for campus placements. Students generally use different websites and applications for aptitude preparation, reasoning, English, programming, coding practice, company-specific questions, mock tests, interview preparation, and resume improvement. Managing preparation across multiple platforms can be time-consuming and makes it difficult to track overall progress.

The proposed Placement Preparation Portal solves this problem by bringing all major placement-preparation activities into a single platform. Students can study preparation notes, practice topic-wise questions, attempt quizzes and mock tests, solve coding problems, prepare for specific companies, scan and improve their resumes, and monitor their performance through a personalized dashboard.

The system consists of two major panels:

- Student Panel: Provides access to learning materials, aptitude, reasoning, English, coding practice, mock tests, company preparation, Resume Scanner, AI Insights, and Analytics.
- Admin Panel: Allows administrators to manage students, notes, questions, coding problems, companies, mock tests, results, and other preparation content dynamically.
-

The project is developed using the MERN Stack, consisting of MongoDB, Express.js, React.js, and Node.js, along with technologies such as Mongoose, JWT, bcryptjs, Axios, React Router, and Tailwind CSS.

The main purpose of the project is to create a single, dynamic, user-friendly, and data-driven placement preparation system where students can follow the complete preparation cycle:

Learn → Practice → Test → Analyze → Improve

Through progress tracking and result analysis, students can identify their strengths and weaknesses and improve their placement readiness systematically.

2. PREAMBLE

2.1 Synopsis :

The **Placement Preparation Portal** is a comprehensive, web-based student placement preparation system designed to provide all essential placement preparation resources through a **single centralized platform**. The main purpose of the system is to help students prepare systematically for different stages of campus recruitment, including aptitude examinations, reasoning tests, English assessments, technical coding rounds, company-specific examinations, technical interviews, HR interviews, and resume evaluation.

In the traditional placement preparation process, students generally depend on multiple websites, YouTube videos, books, coding platforms, and other resources. This makes preparation difficult to organize and makes it challenging for students to determine their actual preparation level. The Placement Preparation Portal addresses these problems by integrating different preparation activities into one unified web application.

The system follows a complete preparation cycle:

Learn → Practice → Quiz → Mock Test → Analyze Result → Identify Weak Areas → Improve

The application contains two major user panels: **Student Panel** and **Admin Panel**. Each panel has a different purpose and role-based access.

2.2 Problem Statement :

Students preparing for campus placements generally depend on multiple websites and applications for different preparation requirements such as aptitude, reasoning, English, programming, coding problems, mock tests, company-specific questions, interview preparation, and resume improvement.

This fragmented approach creates several problems:

- Difficulty in finding all preparation material in one place.
- Lack of structured topic-wise preparation.
- Difficulty in tracking individual progress.
- Limited performance analysis.
- Separate platforms for coding, aptitude, reasoning, and English.
- Difficulty in preparing according to specific companies.
- Lack of centralized mock-test results.
- Difficulty in identifying strong and weak areas.
-

Therefore, there is a need for a **single integrated placement preparation platform** that provides students with learning, practice, testing, analysis, and career preparation facilities.

2.3 Objectives of the System :

The main objectives of the **Placement Preparation Portal** are:

1. Develop a Centralized Platform for Student Placement Preparation

The system provides a single platform where students can prepare for different placement stages.

It brings aptitude, reasoning, English, coding, interviews, and company preparation together.

2. Provide Topic-Wise Preparation Notes and Study Material

Students can access organized notes and learning materials for individual topics.

This allows students to understand concepts before attempting practice questions and tests.

3. Provide Aptitude, Reasoning, and English Practice Questions

The portal provides subject-wise and topic-wise questions for regular practice.

Students can solve MCQs, quizzes, and practice sets to improve their examination performance.

4. Provide Programming Language and Coding Preparation

Students can learn important programming languages and technical concepts required for placement interviews.

The system provides programming theory, examples, interview questions, and coding-related practice.

5. Provide Coding Challenges with Different Difficulty Levels

The Coding Practice module provides coding problems categorized as **Easy, Medium, and Hard**.

This enables students to gradually improve their problem-solving and programming skills.

6. Conduct Quizzes and Timed Mock Placement Tests

Students can participate in topic-wise quizzes and complete placement-style mock examinations.

Timers, question navigation, auto-submission, and scoring make the tests similar to real examinations.

7. Provide Company-Specific Placement Preparation

The portal provides preparation resources based on individual companies and their recruitment patterns.

Students can study selection processes, aptitude, coding, technical, and HR interview questions.

8. Provide a Resume Scanner for Resume Review and Improvement

Students can upload their resumes and receive resume-related analysis and improvement suggestions.

The module helps students identify areas that may need improvement before applying for placements.

2.4 Scope of the System :

A. In-Scope Modules & Features

The following modules and features are included in the proposed Placement Preparation Portal:

1. Student Authentication & Profile

- Student registration and login.
- JWT-based authentication and role-based access.
- Profile management, password change, and logout.

2. Student Dashboard

- Placement readiness score.
- Daily and overall progress.
- Study streak and completed topics.
- Recent activity, bookmarks, recommendations, and notifications.
- Performance charts and preparation overview.

3. Aptitude Preparation

- Topic-wise notes and study material.
- MCQs and practice questions.
- Topic-wise quizzes.
- Timed mock tests.
- Test results and performance analysis.
- Topics such as Number System, Percentage, Profit & Loss, Average, Ratio, Time & Work, Probability, SI, CI, and others.

4. Reasoning Preparation

- Logical and analytical reasoning notes.
- Topic-wise examples and practice questions.
- MCQs and quizzes.
- Mock tests with timer.
- Results, explanations, bookmarks, and progress tracking.

5. English Preparation

- Grammar and vocabulary preparation.
- Synonyms, antonyms, idioms, and one-word substitution.
- Reading comprehension.
- Sentence correction and error detection.
- Quizzes, mock tests, results, and progress tracking.

6. Programming Languages

- Preparation material for C, C++, Java, Python, JavaScript, PHP, SQL, HTML, CSS, React, Node.js, Express.js, and MongoDB.
- Programming concepts, syntax, examples, programs, MCQs, and interview questions.

7. Coding Practice

- Data Structures and Algorithms preparation.
- Arrays, Strings, Stack, Queue, Linked List, Trees, Graphs, Sorting, Searching, Recursion, and Dynamic Programming.
- Coding challenges categorized into Easy, Medium, and Hard.
- Hints, solutions, explanations, bookmarks, search, filters, and progress tracking.

8. Company Preparation

- Company-specific placement preparation.
- Selection process information.
- Aptitude, coding, technical, and HR questions.
- Interview experiences and preparation tips.
- Company-wise search and filtering.

9. Mock Test System

- Aptitude, Reasoning, English, Coding, and Mixed Placement Tests.
- Timer and question navigation.
- Auto-submit and negative marking.
- Score calculation and detailed result analysis.
- Test history and performance tracking.

10. Resume Scanner

- Resume upload and storage.
- Resume review and analysis.
- Suggestions for improving resume content.
- Identification of potential missing or weak areas.
- AI-assisted resume analysis where configured.

11. AI Preparation

- AI-generated HR interview questions.
- Technical interview question generation.
- Resume suggestions.
- Skill-gap analysis.
- Personalized learning roadmap.
- Mock interview feedback.
- Career-related suggestions.

12. Progress & Analytics

- Daily, weekly, and monthly progress.
- Topic-wise performance.
- Coding performance.
- Mock test history.
- Accuracy and scores.
- Strong and weak areas.
- Placement readiness tracking.

13. Admin Panel

- Admin authentication and dashboard.
- Student management.
- Notes management.
- Aptitude, Reasoning, and English question banks.
- Coding problem management.
- Company management.
- Mock test management.
- Results and attempts management.
- Reports and analytics.
- Dynamic CRUD operations.
-

14. Database & API Integration

- MongoDB-based data storage.
- Mongoose database models.
- REST APIs using Node.js and Express.js.
- Frontend-backend integration using Axios.
- Dynamic content delivery from Admin Panel to Student Panel.

B. Out-of-Scope (Future Boundaries)

The following features are outside the primary scope of the current system and may be considered for future enhancement:

1. Online Compiler Infrastructure

A fully secure online code execution environment for running student programs in multiple programming languages is not part of the current core scope.

2. Real-Time AI Voice Interview

Real-time voice-based interviews with speech recognition, voice analysis, and conversational AI can be implemented as a future enhancement.

3. Direct Company Recruitment

The portal does not directly conduct company recruitment, job offers, or hiring processes.

4. Live Placement Drives

Managing real-time campus placement drives, interview scheduling, candidate selection, and offer management is outside the current scope.

5. Mobile Application

A dedicated Android or iOS application is not included in the current web-based implementation and may be developed in the future.

6. Advanced Job Matching

Automatic matching of student profiles with live job vacancies based on skills, experience, and eligibility can be added in a future version.

7. Advanced Resume Parsing

More advanced resume parsing, ATS simulation, job-specific resume optimization, and automated job matching can be added later.

8. External Placement Data Integration

Direct integration with external job portals, company recruitment systems, or third-party placement databases is outside the current scope.

9. Multilingual Learning

Providing complete preparation content in multiple regional languages can be considered as a future enhancement.

10. Advanced Predictive Analytics

Advanced machine-learning models for predicting placement probability, salary range, or company selection probability are outside the current implementation boundary.

Scope Summary

In-Scope:

Learn → Practice → Quiz → Mock Test → Result → Analysis → Improvement

Future Scope:

AI Voice Interview → Online Compiler → Job Matching → Mobile App → Advanced Predictive Analytics → Live Recruitment Integration

2.5 Module description :

1. Authentication Module

This module provides secure registration and login for Students and Admins. It includes JWT authentication, password management, profile updates, and role-based access control.

2. Student Dashboard Module

This module provides students with an overview of their placement preparation and performance. It displays progress, readiness score, study streak, completed topics, activities, and recommendations.

3. Aptitude Module

This module helps students prepare for quantitative aptitude through topic-wise notes, examples, MCQs, quizzes, and mock tests. It also provides timers, results, explanations, and performance tracking.

4. Reasoning Module

This module provides preparation for logical and analytical reasoning topics commonly asked in placement examinations. Students can study notes, practice questions, attempt quizzes and mock tests, and track their progress.

5. English Module

This module improves students' English and verbal ability through grammar, vocabulary, comprehension, sentence correction, and other placement-related topics. It provides study material, practice questions, quizzes, and mock tests.

6. Programming Languages Module

This module provides theoretical and practical preparation for important programming languages and technologies. Students can learn concepts, syntax, examples, programs, MCQs, and interview questions.

7. Coding Practice Module

This module helps students improve programming and problem-solving skills through Data Structures and Algorithms. It provides coding challenges categorized into Easy, Medium, and Hard levels along with solutions, hints, explanations, and progress tracking.

8. Company Preparation Module

This module provides company-specific placement preparation resources. Students can study selection processes, aptitude, coding, technical and HR questions, interview experiences, and preparation tips.

9. Mock Test Module

This module allows students to take placement-oriented tests for Aptitude, Reasoning, English, Coding, and Mixed subjects. It includes timers, question navigation, auto-submit, negative marking, scoring, and detailed result analysis.

10. Resume Scanner Module

This module allows students to upload their resumes and receive resume-related analysis and improvement suggestions. It is designed to help students identify missing information and improve the quality of their resumes.

11. AI Preparation Module

This module provides AI-assisted placement preparation such as HR and technical interview questions, resume suggestions, skill-gap analysis, learning roadmaps, and interview feedback.

12. Progress & Analytics Module

This module analyzes student preparation and displays daily, weekly, and monthly performance. It tracks accuracy, completed topics, coding progress, mock-test history, strengths, weaknesses, and placement readiness.

13. Admin Module

This module allows administrators to manage the complete portal and its content. Admins can manage students, notes, questions, coding problems, companies, mock tests, results, reports, and analytics through dynamic CRUD operations.

3. REVIEW OF LITERATURE

The Placement Preparation Portal is based on the study of existing online learning, coding practice, aptitude preparation, mock-test, resume analysis, and interview preparation platforms. The review of existing systems helps identify their useful features as well as the limitations that can be addressed through an integrated placement preparation platform.

3.1 Online Learning Platforms

Online learning platforms provide students with structured courses, study materials, videos, quizzes, and assessments. These platforms demonstrate the importance of organizing educational content according to subjects and topics. However, general learning platforms are not specifically focused on the complete campus placement preparation process.

3.2 Coding Practice Platforms

Coding practice platforms provide programming problems based on different topics and difficulty levels. Features such as problem-solving, hints, solutions, explanations, and progress tracking are useful for technical placement preparation. The proposed system incorporates similar coding-practice concepts while combining them with aptitude, reasoning, English, company preparation, and mock tests.

3.3 Aptitude and Reasoning Preparation Systems

Aptitude and reasoning preparation websites generally provide topic-wise questions, MCQs, explanations, quizzes, and practice tests. These systems are useful for preparing for the assessment rounds of companies. The proposed portal integrates these preparation areas into the same platform and connects their results with the student's overall progress.

3.4 Mock Test Systems

Online examination systems provide features such as timers, question navigation, automatic submission, scoring, and result analysis. These features are important for simulating real placement examinations. The Placement Preparation Portal extends this concept by providing aptitude, reasoning, English, coding, and mixed placement mock tests.

3.5 Company-Specific Preparation

Company preparation platforms provide information about recruitment processes, frequently asked questions, coding problems, aptitude questions, and interview experiences. Such resources help students prepare according to specific company requirements. The proposed system includes a dedicated Company Preparation module to organize this information in one place.

3.6 Resume Analysis Systems

Resume-related systems help candidates create, review, or improve their resumes. Resume analysis can identify areas such as missing information, skills, formatting, and content quality. The proposed Resume Scanner module incorporates resume upload and review-oriented functionality and can be extended with AI-based suggestions.

3.7 AI-Based Interview Preparation

Recent AI-assisted career tools demonstrate the potential of artificial intelligence for generating interview questions, providing feedback, analyzing skills, and recommending learning paths. The proposed system includes an AI module for HR and technical interview questions, resume suggestions, skill-gap analysis, learning roadmaps, and interview feedback.

3.8 Research Gap

The review indicates that existing systems generally focus on a particular area such as coding, aptitude, online learning, mock testing, or resume preparation. Students may therefore need to use several different platforms for complete placement preparation.

The Placement Preparation Portal attempts to address this gap by integrating multiple preparation activities into one platform:

Study Material → Practice → Quiz → Mock Test → Result Analysis → Progress Tracking → Improvement

4. TECHNICAL DESCRIPTION

4.1 Hardware Requirement

This section specifies the empirical technical requirements verified directly from the **MARKETA** codebase (React 19.2.7, Vite 8.1.1, React-to-Print 3.3.0, Recharts 3.10.1, Axios 1.18.1, Inter Google Font, and responsive CSS layout engines) needed to open, run, and operate the web application on a user terminal.

Hardware Component	Minimum System Requirement	Recommended System Requirement
Device Category	Desktop PC, Laptop, POS Terminal, or Tablet	Desktop PC or Dedicated POS Touch Terminal
Processor (CPU)	Intel Core i3 / AMD equivalent	Intel Core i5 or higher
System Memory (RAM)	3 GB DDR3	8 GB or higher
Storage	10 GB free space	20 GB or more
Display Resolution	1366 × 768 resolution	Full HD 1920 × 1080
Internet	Basic broadband connection	Stable high-speed internet
Keyboard	Standard keyboard	Standard keyboard
Mouse	Standard mouse	Standard mouse

Table 4.1 Hardware Requirement

4.2 Software Requirement

The Placement Preparation Portal requires the following software and development technologies:

Software / Technology	Purpose
Windows / Linux / macOS	Operating System
Node.js	Backend runtime environment
React.js	Frontend development
Vite	React development and build tool
Tailwind CSS	User interface styling
React Router	Frontend navigation and routing
Express.js	Backend API development
MongoDB	Database management
Mongoose	MongoDB object modeling
JWT	Authentication and authorization
bcryptjs	Password hashing
Axios	Client-server API communication
Visual Studio Code	Development environment
Git/GitHub	Version control and source-code management
Postman	API testing
Web Browser	Application access and testing

Table 4.2 Software Requirement

Network & Connectivity Requirements :

The Placement Preparation Portal is a web-based application that requires a stable internet connection for communication between the Student/Admin frontend, backend server, database, and optional external services.

1. Internet Connection

A stable internet connection is required to access the portal, load preparation material, attempt tests, submit answers, and view results.

2. Recommended Internet Speed

A minimum internet speed of approximately 2 Mbps is recommended for normal usage. A faster connection provides better performance when loading large content or using AI-related features.

3. Client-Server Communication

The React frontend communicates with the Node.js and Express.js backend through HTTP/HTTPS REST APIs. Axios is used to send requests and receive responses from the server.

4. Database Connectivity

The backend server requires a network connection to communicate with the MongoDB database. Student profiles, questions, notes, tests, results, progress, and other application data

are stored and retrieved through this connection.

5. HTTPS Security

For production deployment, HTTPS should be used to protect authentication credentials, API requests, and other sensitive information while data is transferred between the client and server.

6. AI and External Services

The optional AI features may require internet connectivity to communicate with an external AI API. Email-based features such as password reset or OTP verification may also require an SMTP service.

7. Local Development Network

During development, the frontend and backend may run on separate local ports, for example:

React/Vite Frontend → localhost → Express Backend → MongoDB

The frontend must be configured with the correct backend API URL.

8. Production Connectivity

After deployment, the system requires:

Student/Admin Browser → Internet → Hosted Frontend → Hosted Backend API → MongoDB

Browser Requirements

The application can be accessed using modern web browsers such as:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

5. SYSTEM DESIGN AND DEVELOPMENT

5.1 Flow Chart:

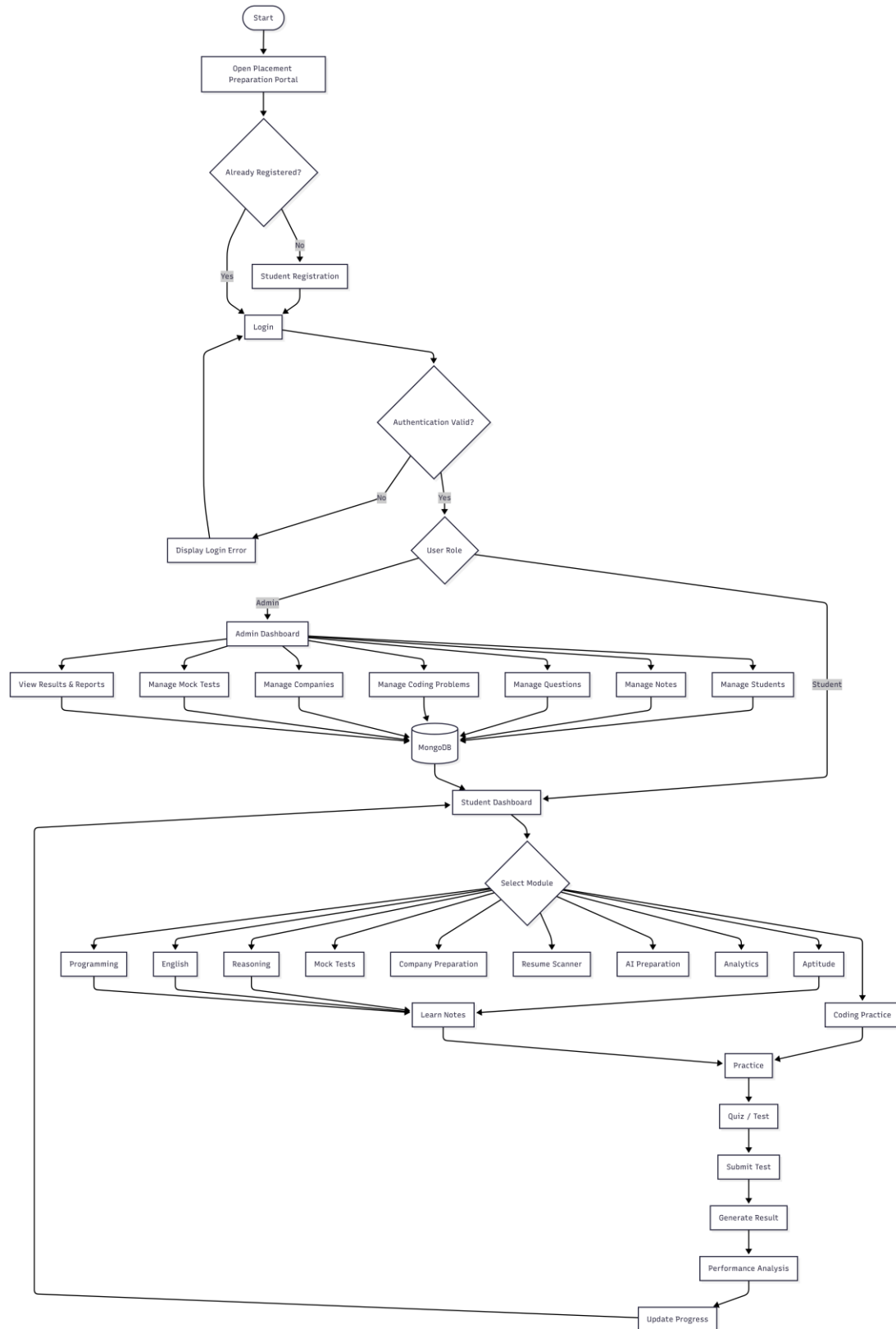


Figure 5.1 Flow Chart

5.2 Data Flow Diagram:

Level 0:-

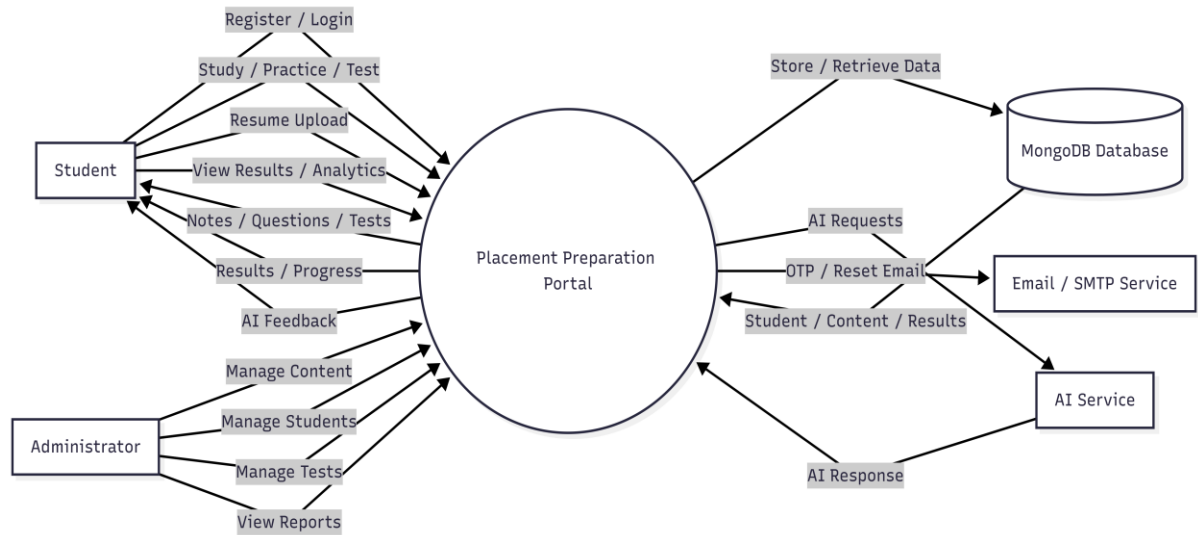


Figure 5.2.1 DFD Level 0

Level 1:-

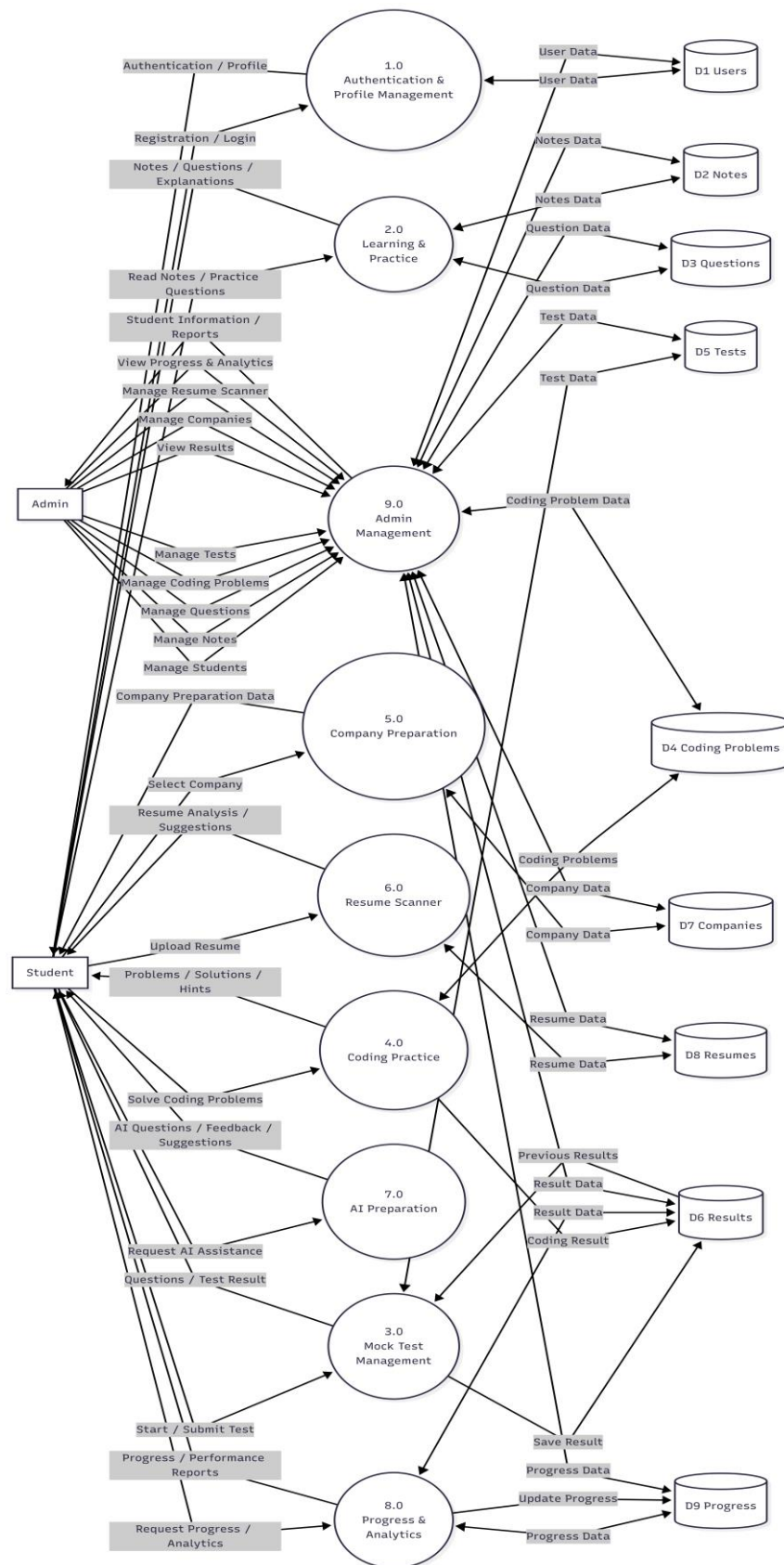


Figure 5.2.2 DFD Level 1

5.3 Use Case Diagram:

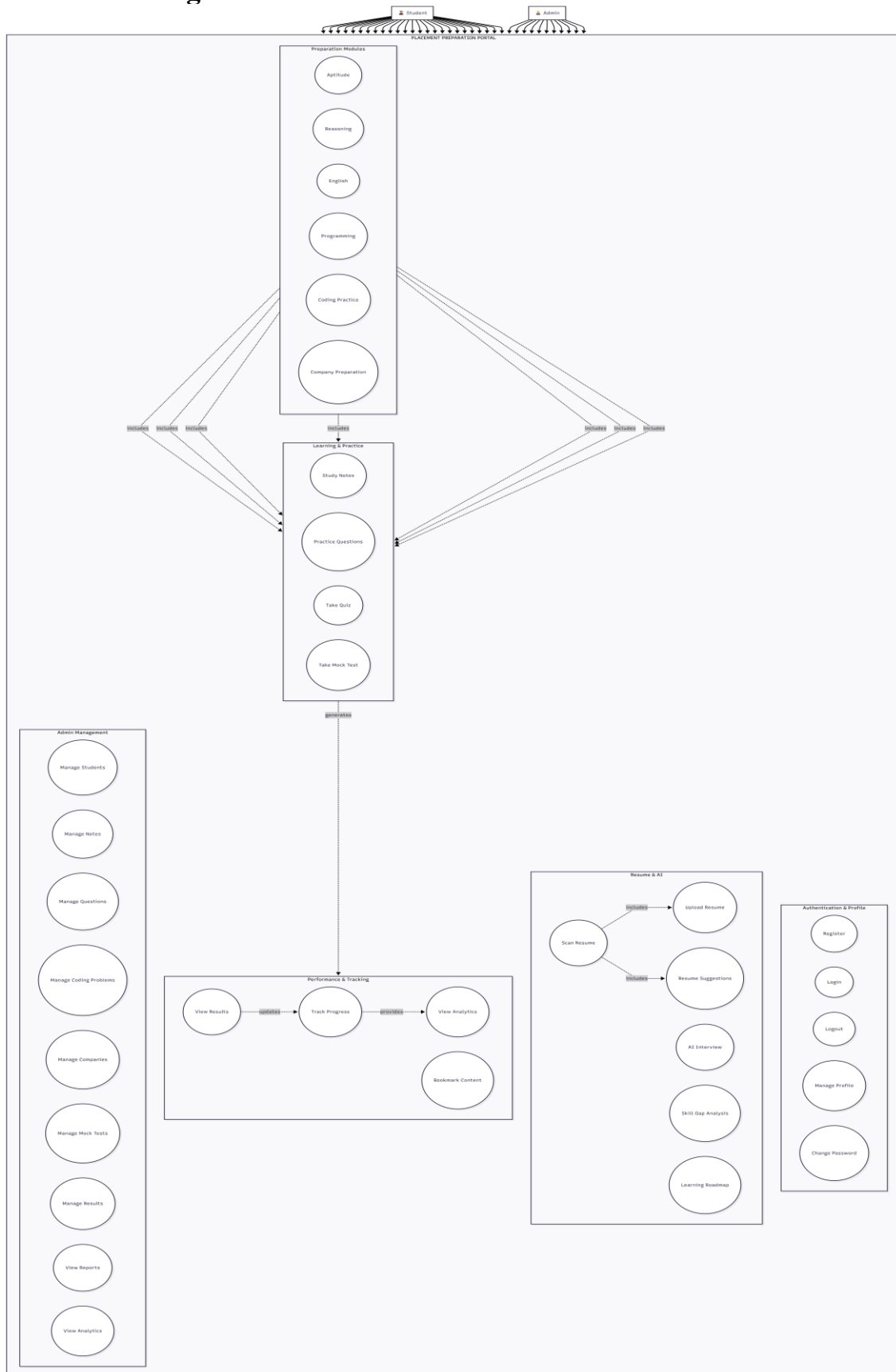


Figure 5.3 Use case diagram

5.4 Sequential Diagram:

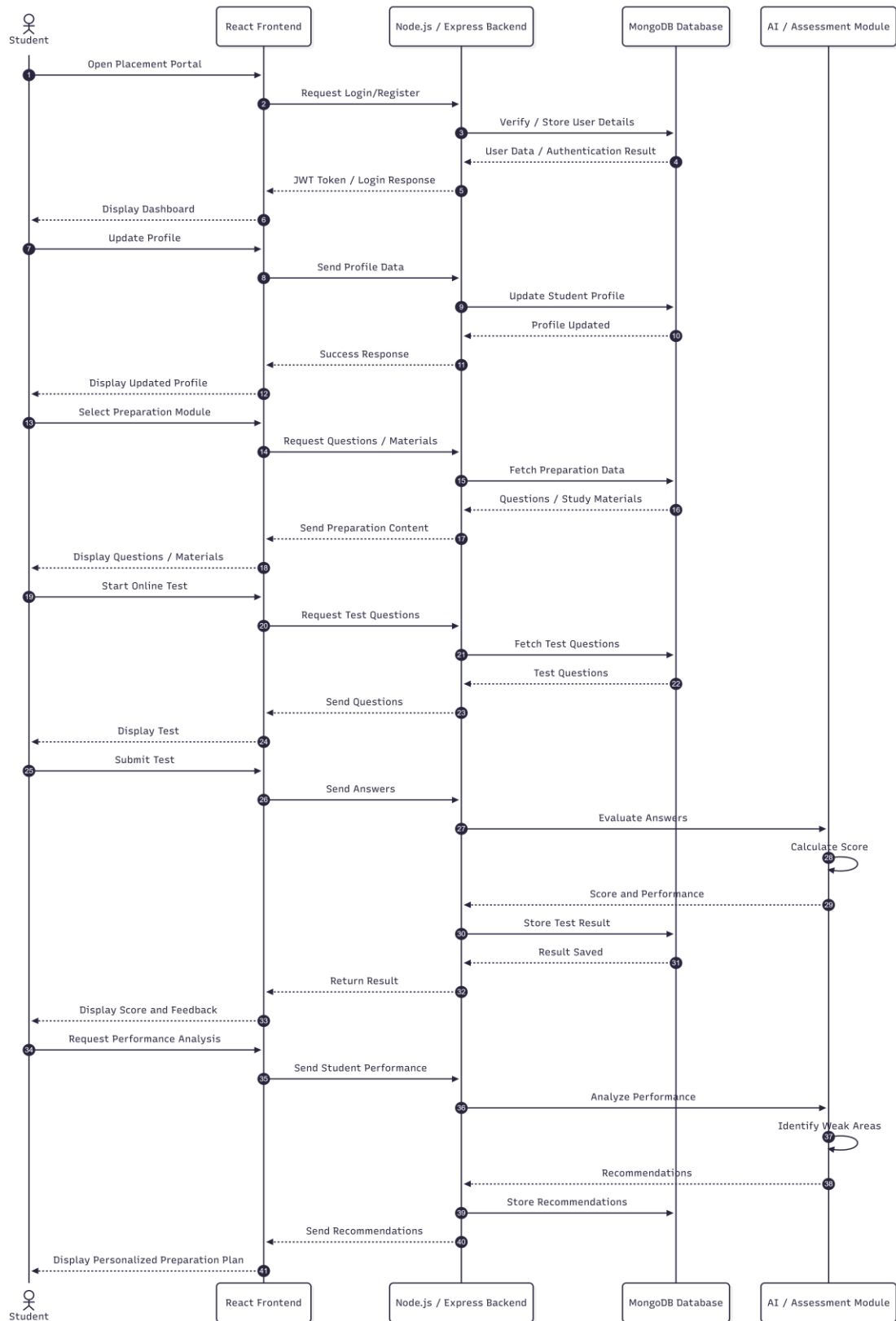


Figure 5.4 Sequential Diagram

5.5 Class Diagram:

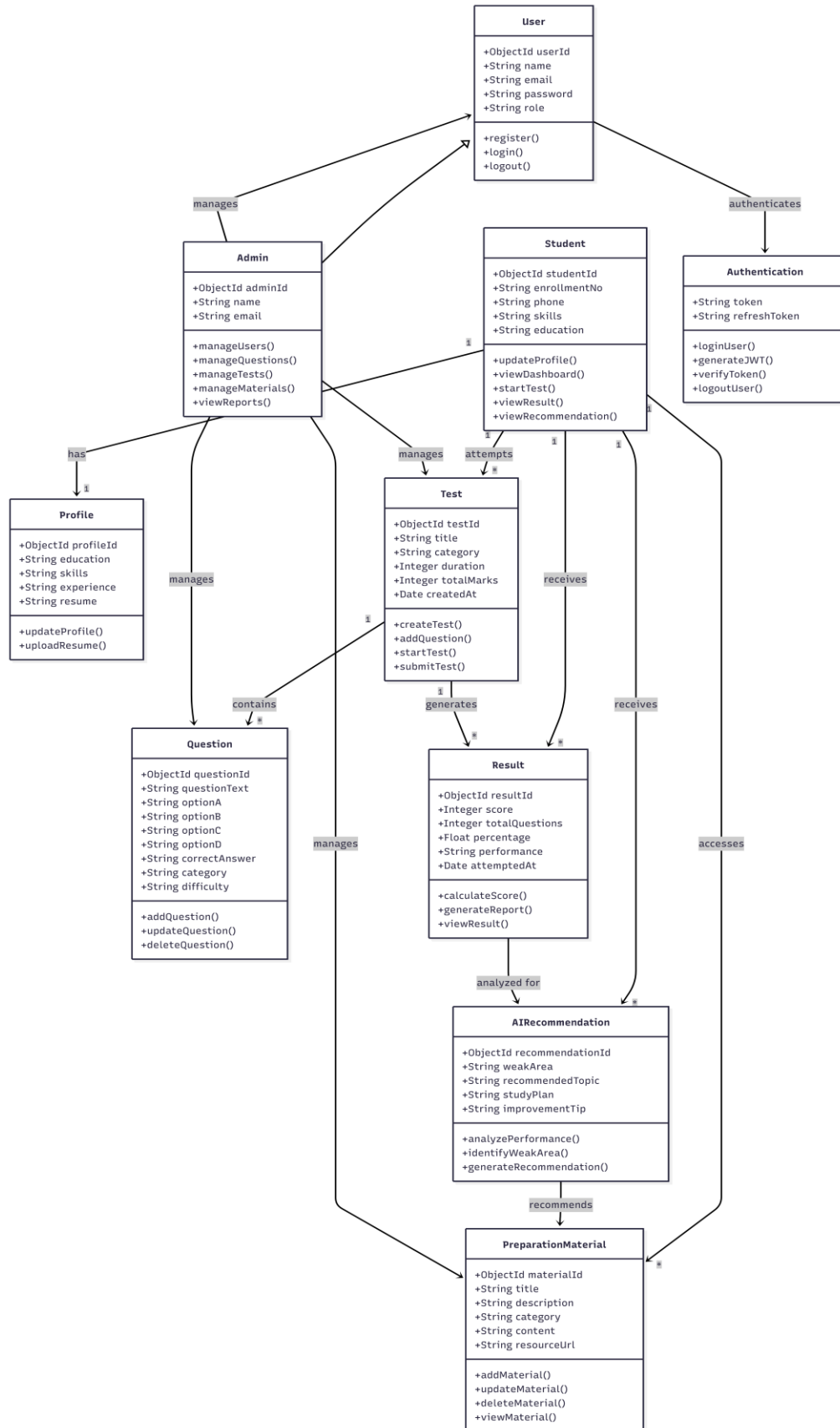


Figure 5.5 Class Diagram

5.6 Activity Diagram:

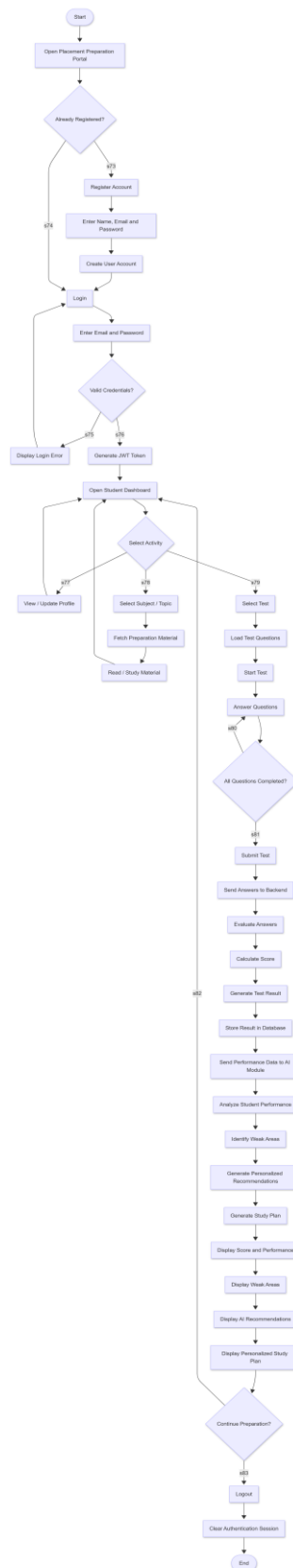


Figure 5.6 Activity Diagram

5.7 Database Design:

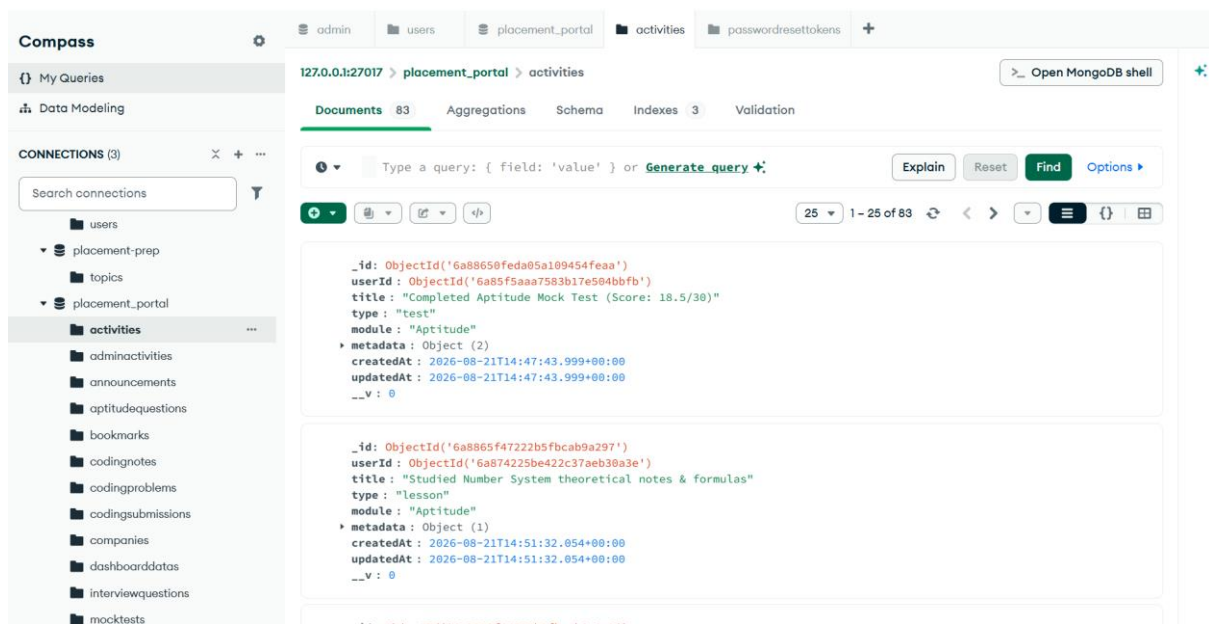


Figure 5.7.1 Activities

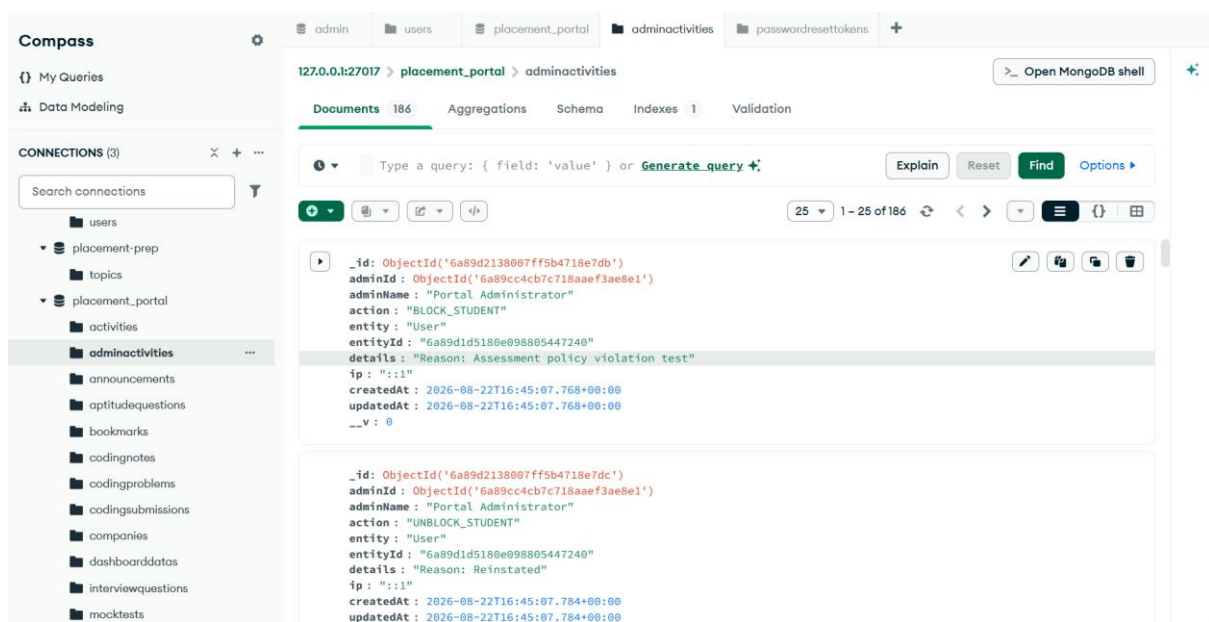


Figure 5.7.2 Adminactivities

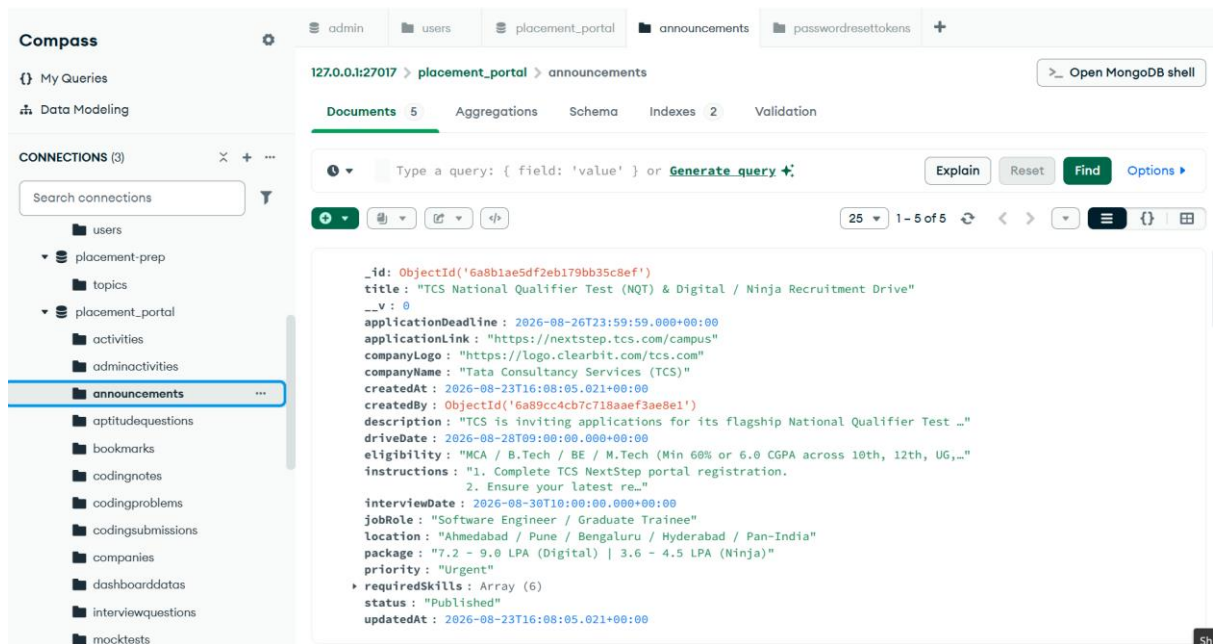


Figure 5.7.3 Announcement

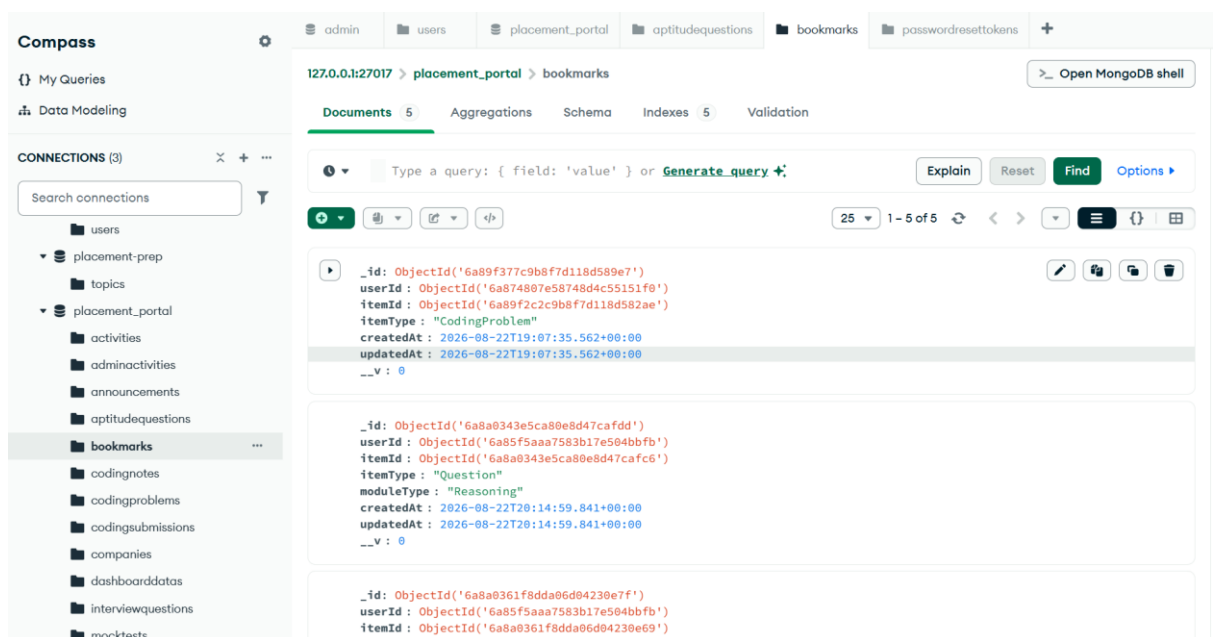


Figure 1.7.4 bookmark

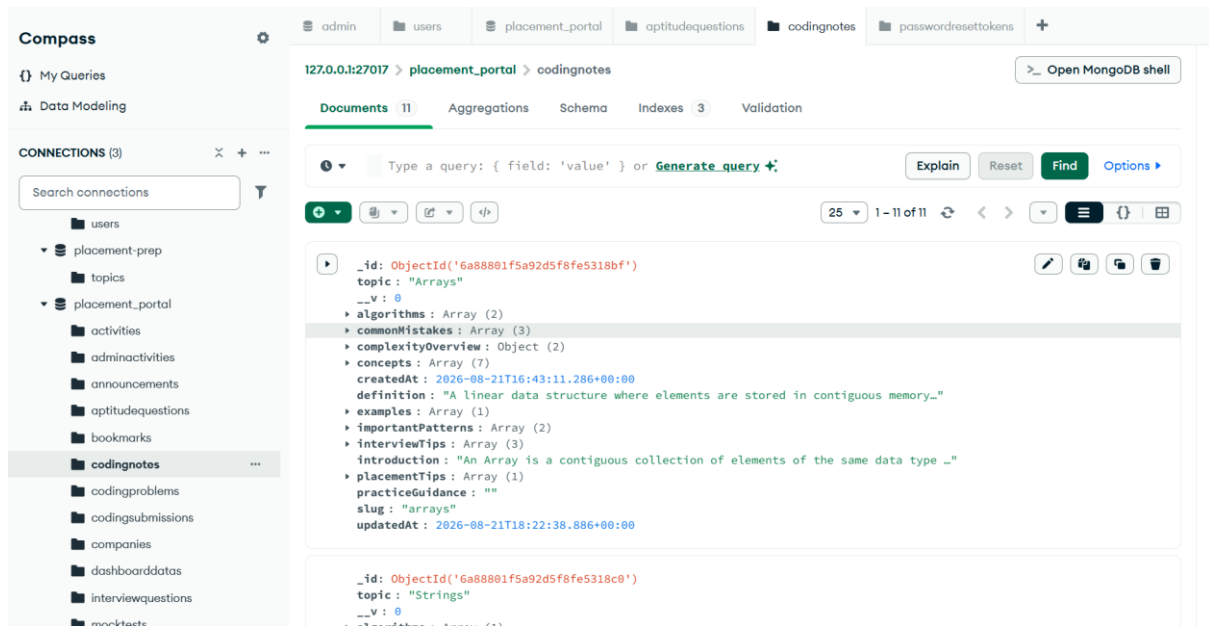


Figure 5.7.5 Codingnotes

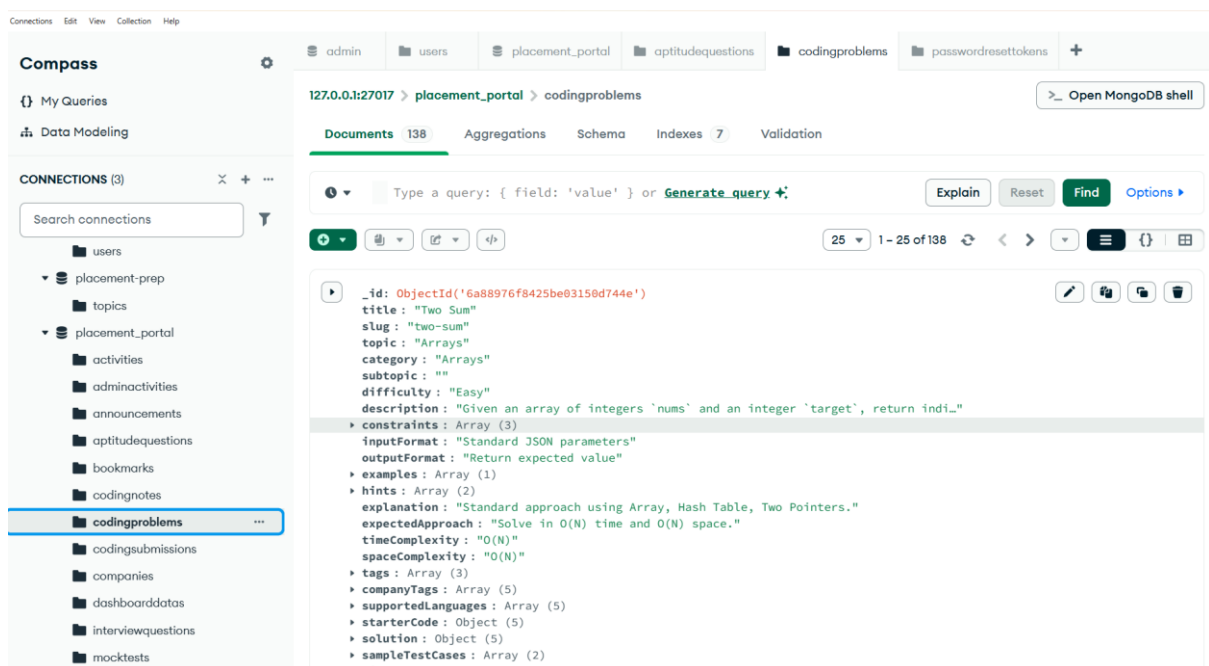


Figure 5.7.6 Codingproblems

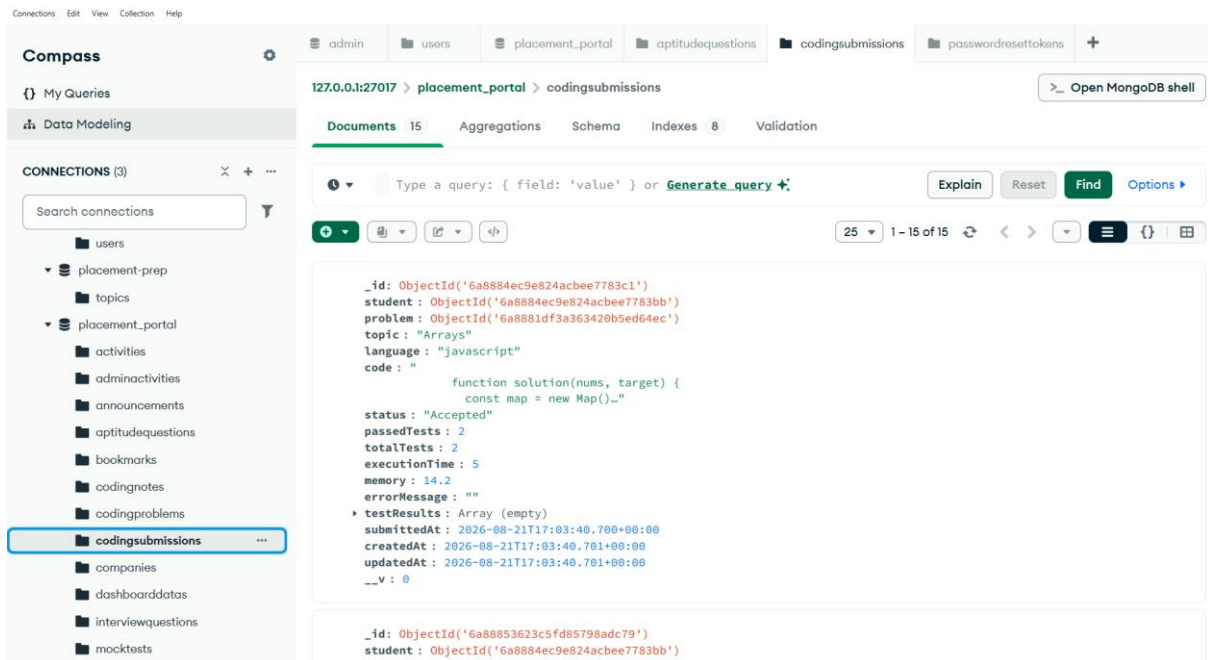


Figure 5.7.7 Codingsubmissionond

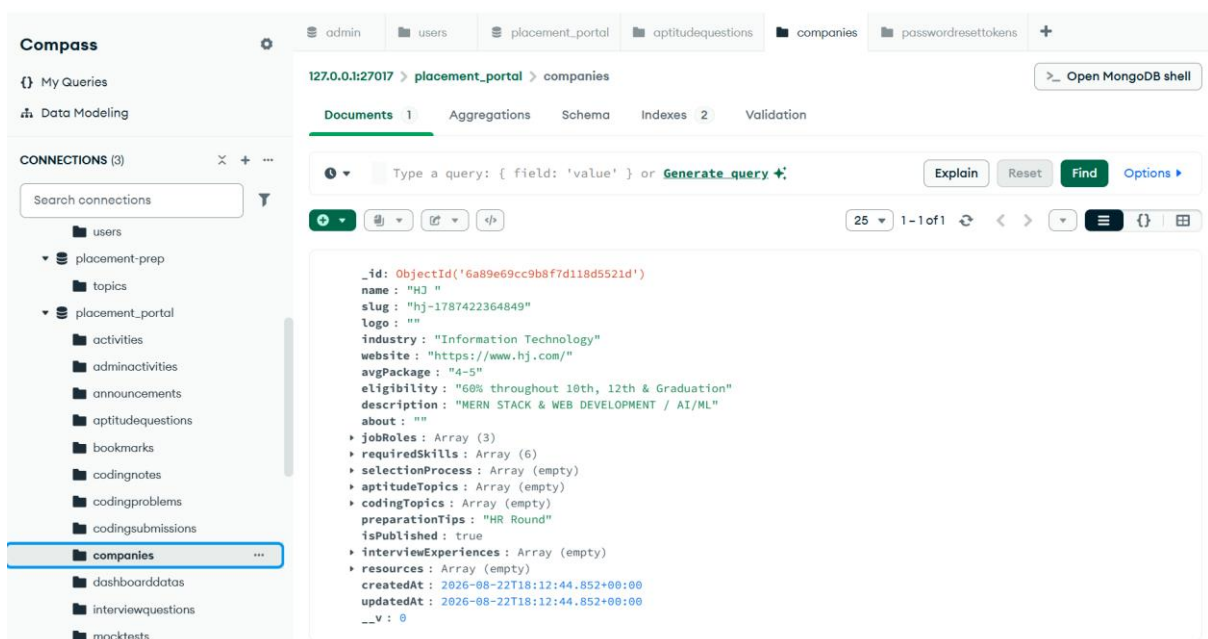


Figure 5.7.8 Companies

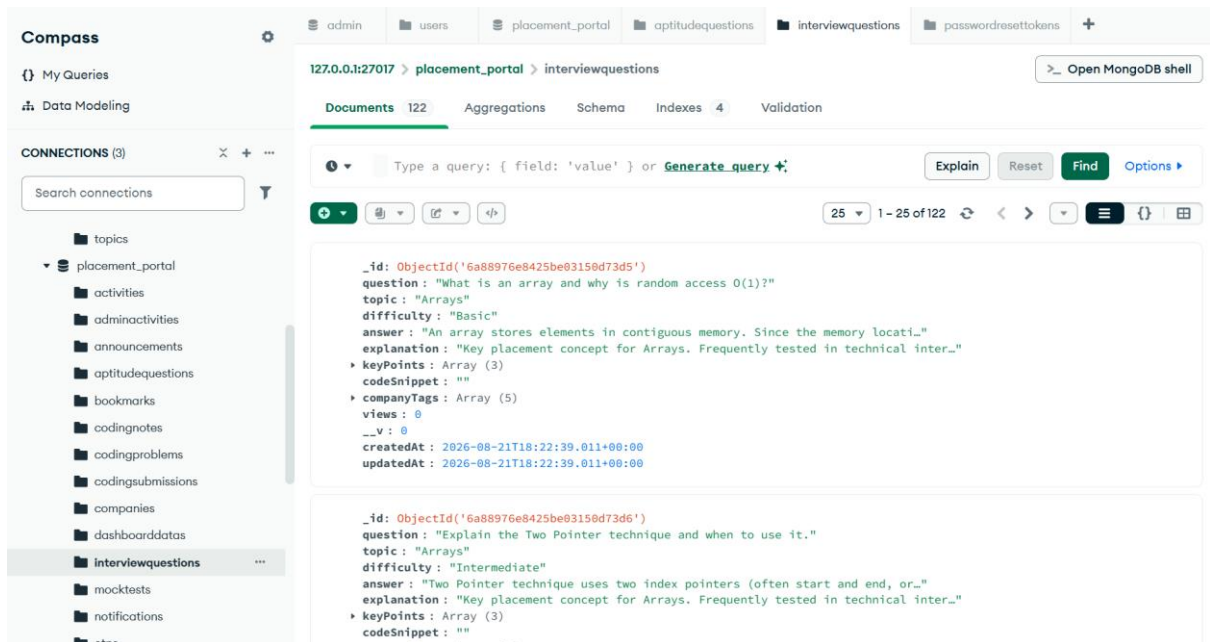


Figure 5.7.9 Interviewquestions

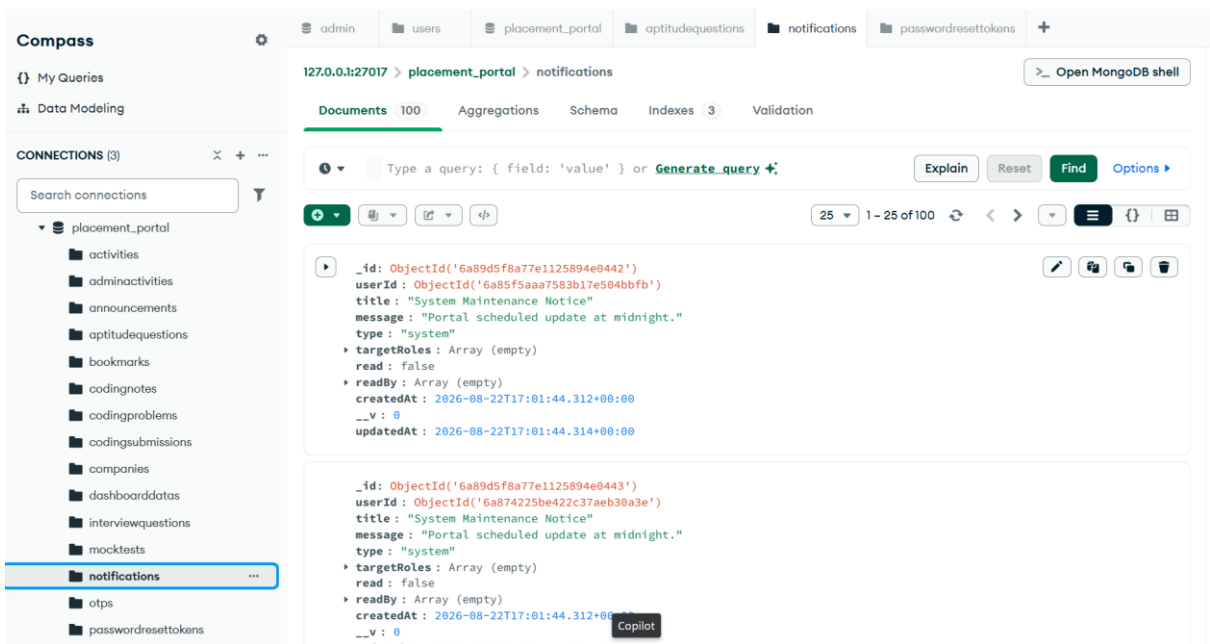


Figure 5.7.10 Notification

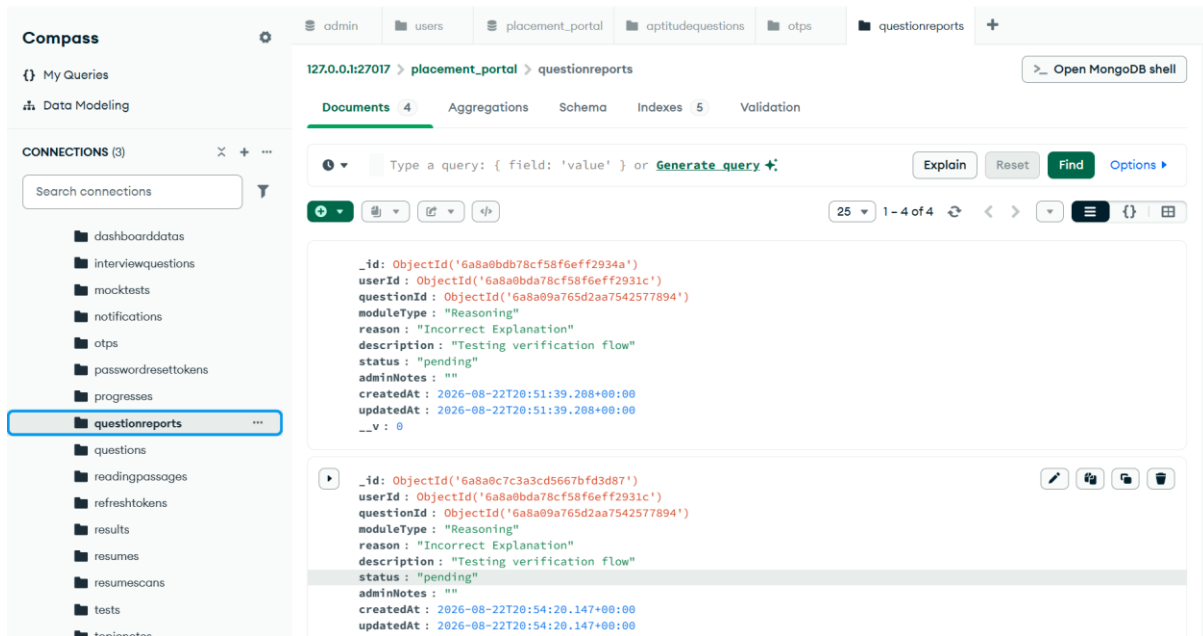


Figure 5.7.11 Question Report

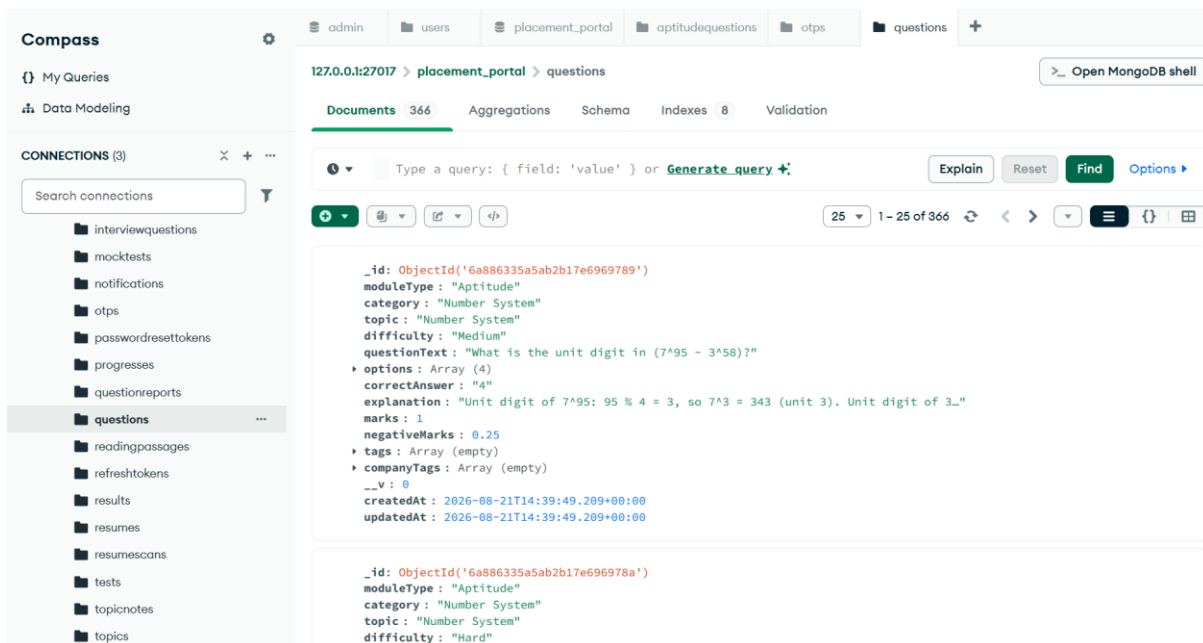


Figure 5.7.12 Questions

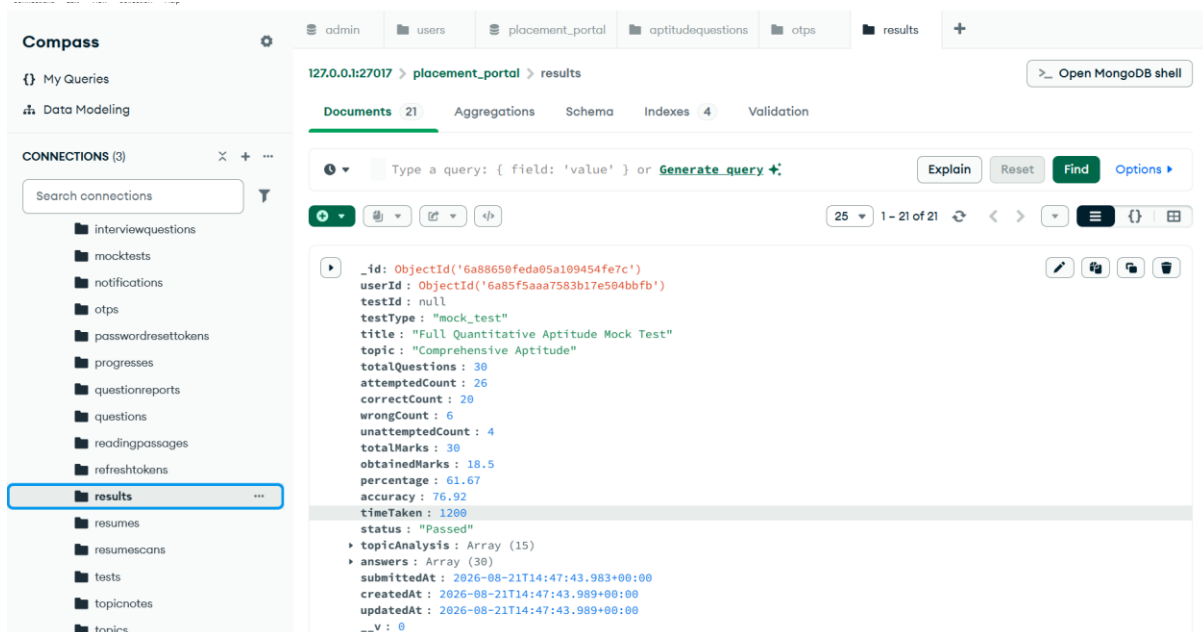


Figure 5.7.13 Result

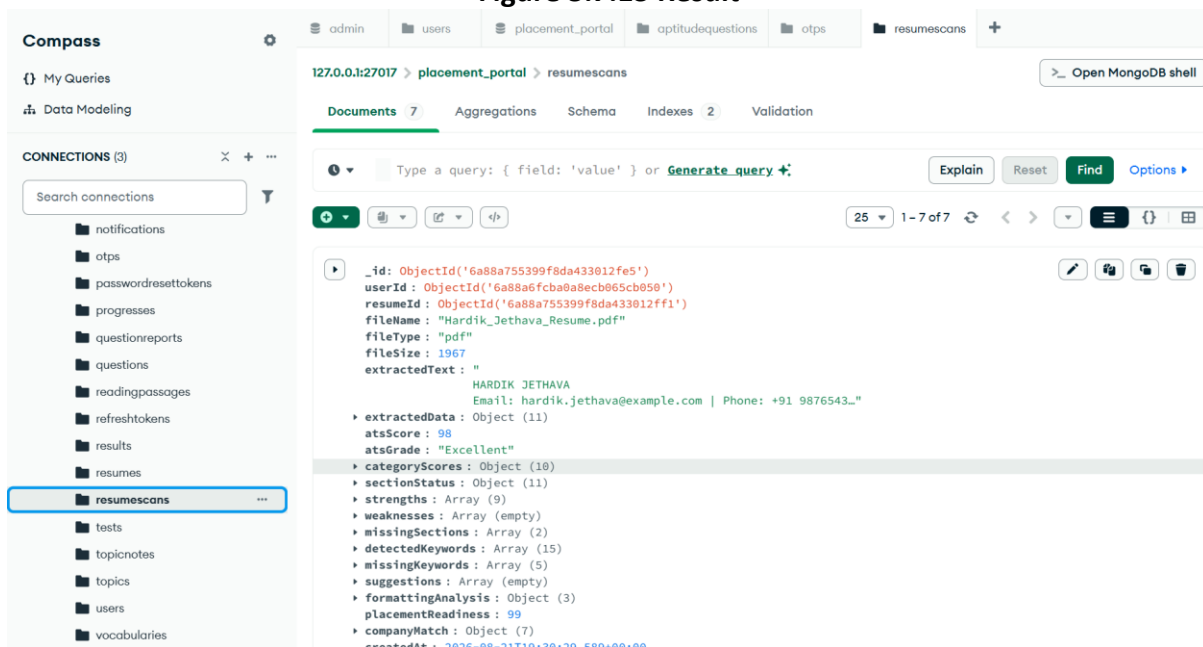


Figure 5.7.14 Resume scans

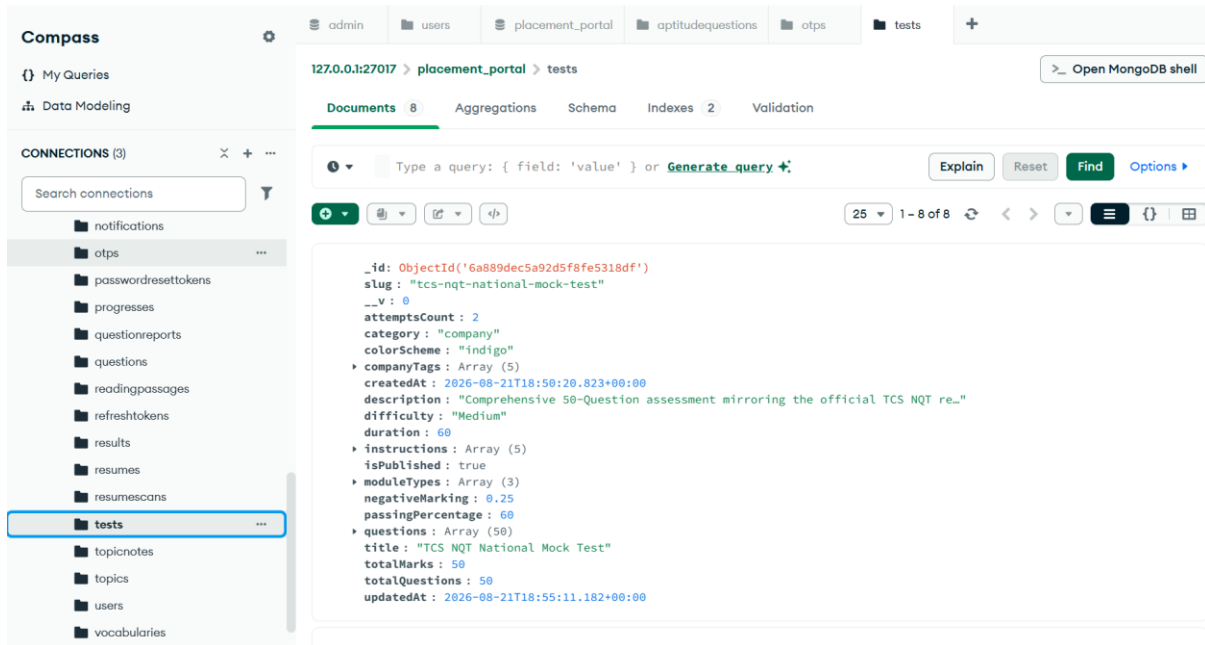


Figure 5.7.15 Test

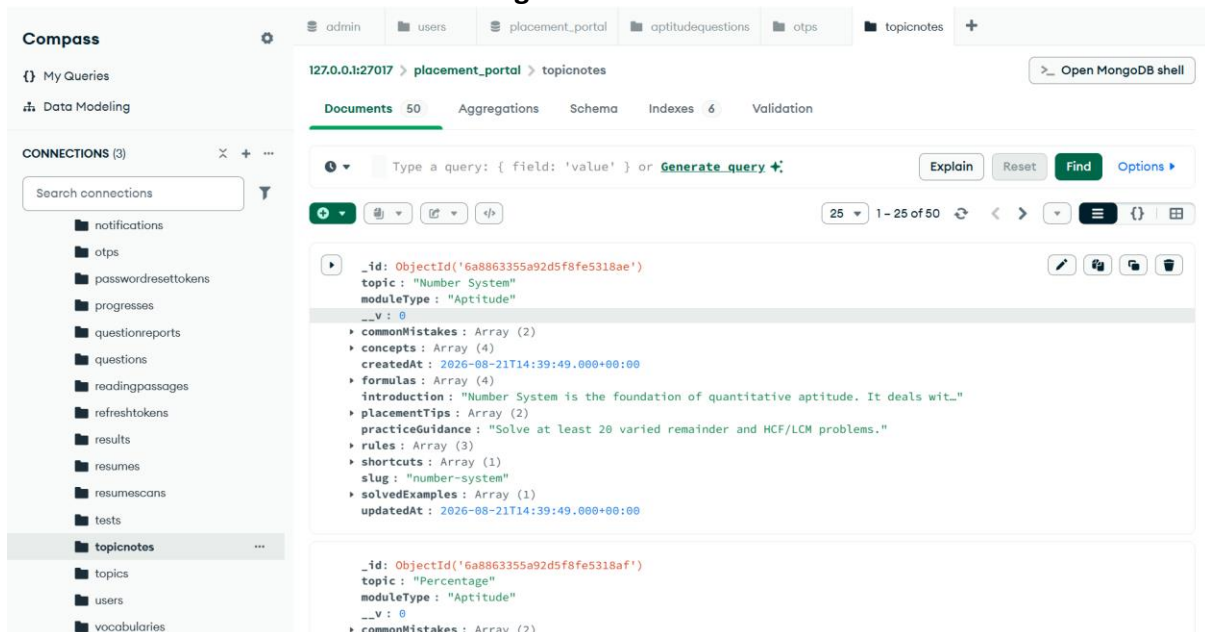


Figure 5.7.16 Topic Notes

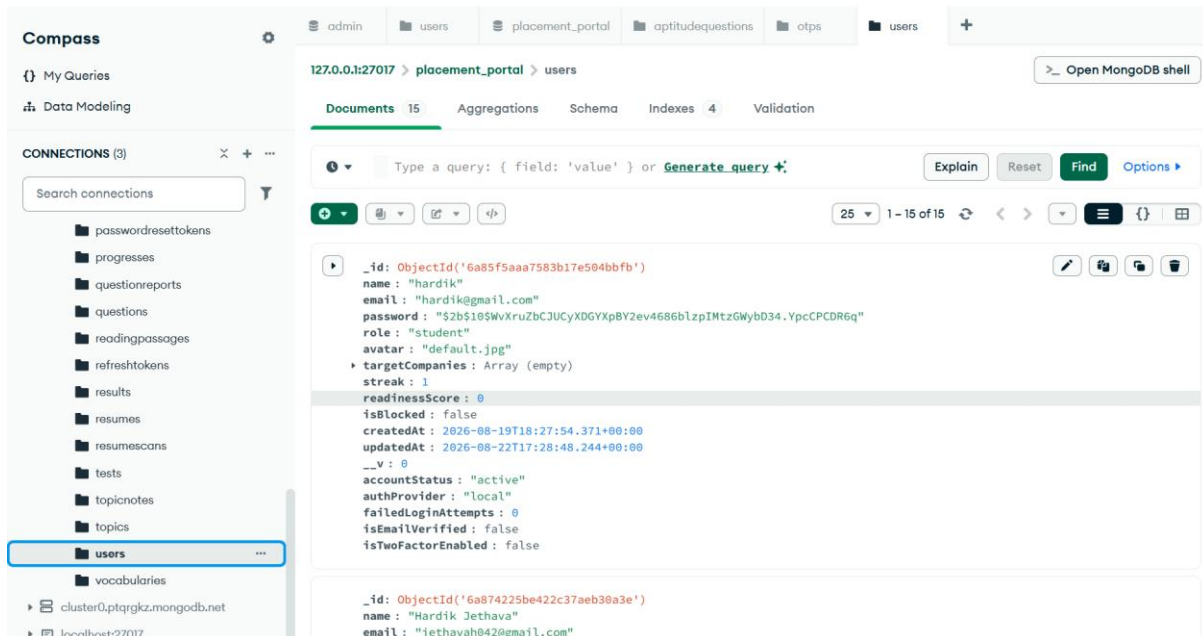


Figure 5.7.17 Users

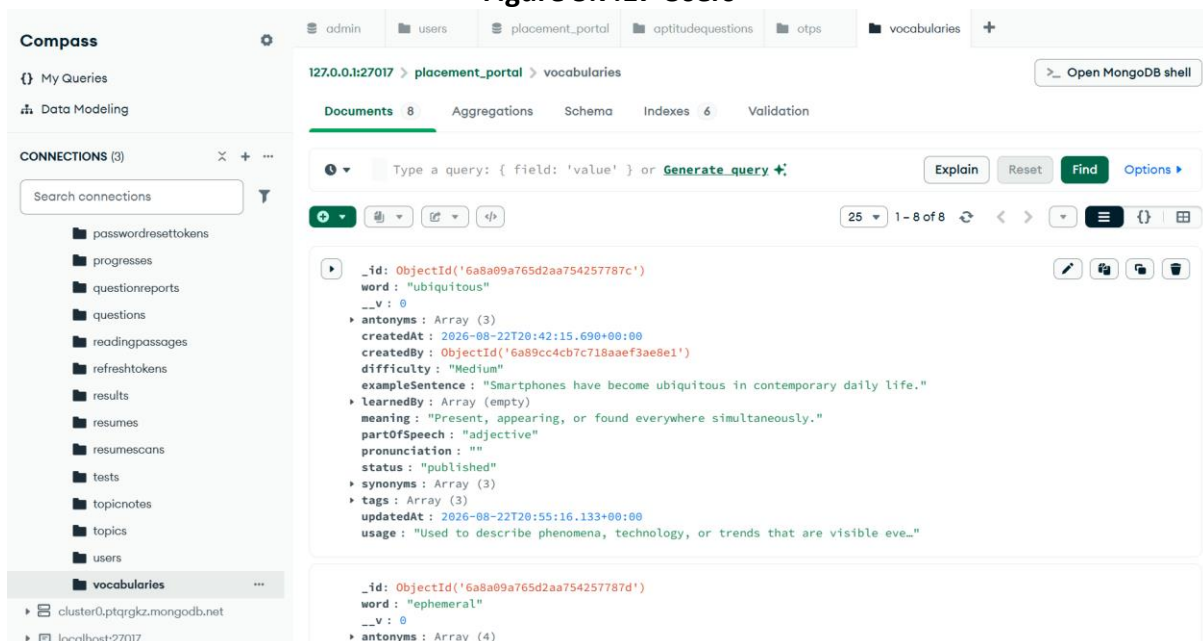
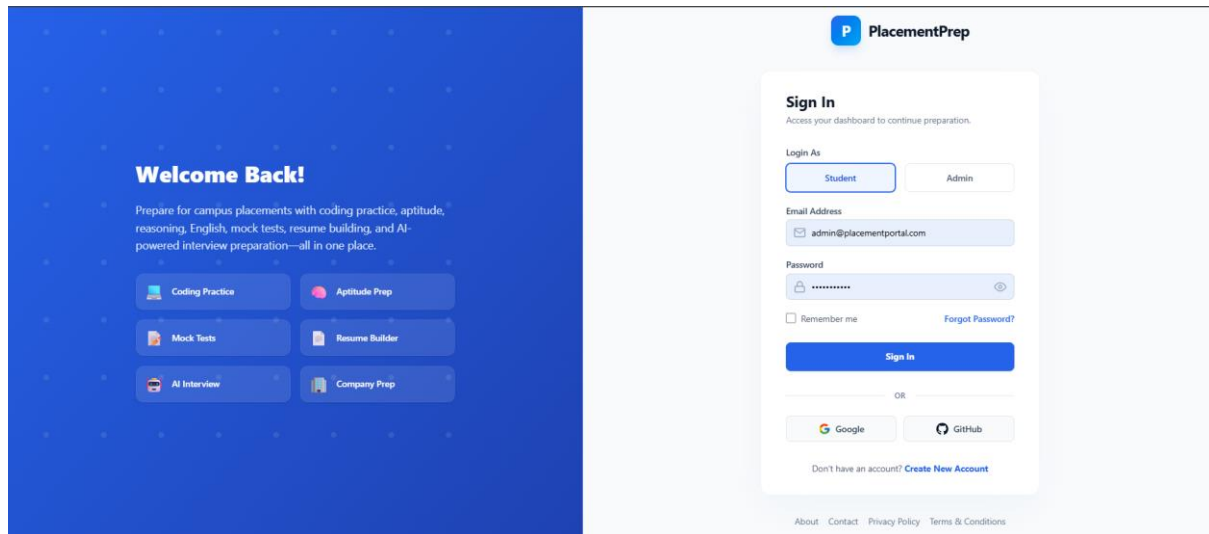


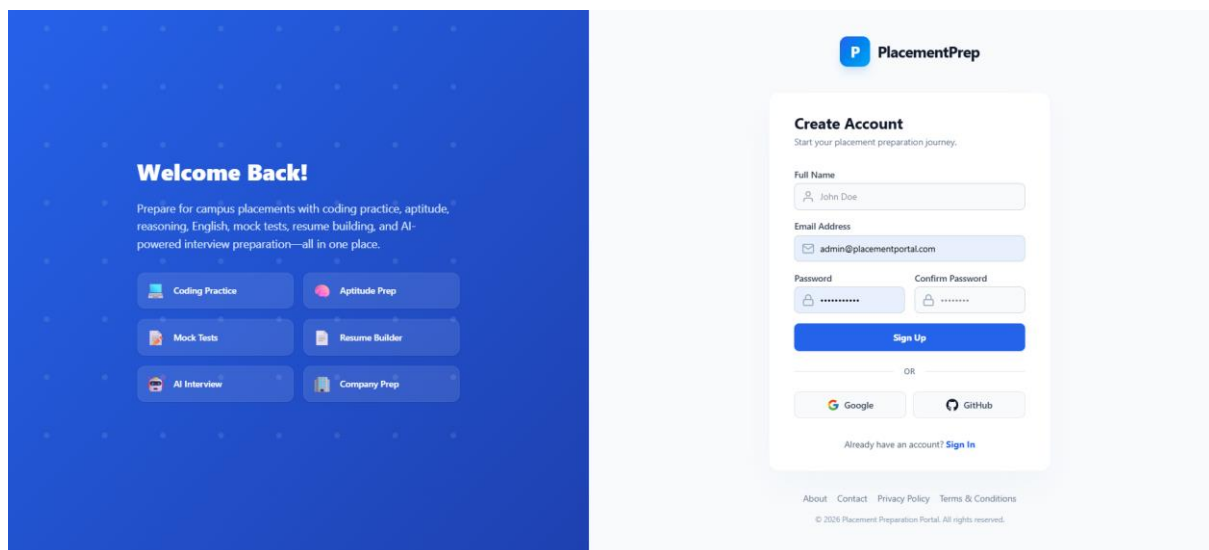
Figure 5.7.18 Vocabulary

5.8 Student Menu-Screen Design:



The figure displays two side-by-side mockups of the PlacementPrep application interface. The left mockup is a 'Welcome Back!' screen with a blue background and a grid of icons for various features: Coding Practice, Aptitude Prep, Mock Tests, Resume Builder, AI Interview, and Company Prep. The right mockup is the 'Sign In' screen, which has a white background and a light blue border. It features the PlacementPrep logo at the top, followed by the 'Sign In' title and a subtitle 'Access your dashboard to continue preparation.' Below this are input fields for 'Login As' (with 'Student' and 'Admin' buttons), 'Email Address' (with the placeholder 'admin@placementportal.com'), and 'Password'. There is a 'Remember me' checkbox and a 'Forgot Password?' link. A blue 'Sign In' button is at the bottom of the form. Below the button is an 'OR' separator and two social login buttons for Google and GitHub. At the very bottom, there is a link 'Don't have an account? Create New Account' and a footer with links for 'About', 'Contact', 'Privacy Policy', and 'Terms & Conditions'.

Figure 5.8.1 Student Login Screen



The figure displays two side-by-side mockups of the PlacementPrep application interface. The left mockup is identical to the one in Figure 5.8.1, showing the 'Welcome Back!' screen with the same blue background and feature icons. The right mockup is the 'Create Account' screen, which has a white background and a light blue border. It features the PlacementPrep logo at the top, followed by the 'Create Account' title and a subtitle 'Start your placement preparation journey.' Below this are input fields for 'Full Name' (with the placeholder 'John Doe'), 'Email Address' (with the placeholder 'admin@placementportal.com'), 'Password', and 'Confirm Password'. A blue 'Sign Up' button is at the bottom of the form. Below the button is an 'OR' separator and two social login buttons for Google and GitHub. At the very bottom, there is a link 'Already have an account? Sign In' and a footer with links for 'About', 'Contact', 'Privacy Policy', and 'Terms & Conditions'.

Figure 5.8.2 Student Register Screen

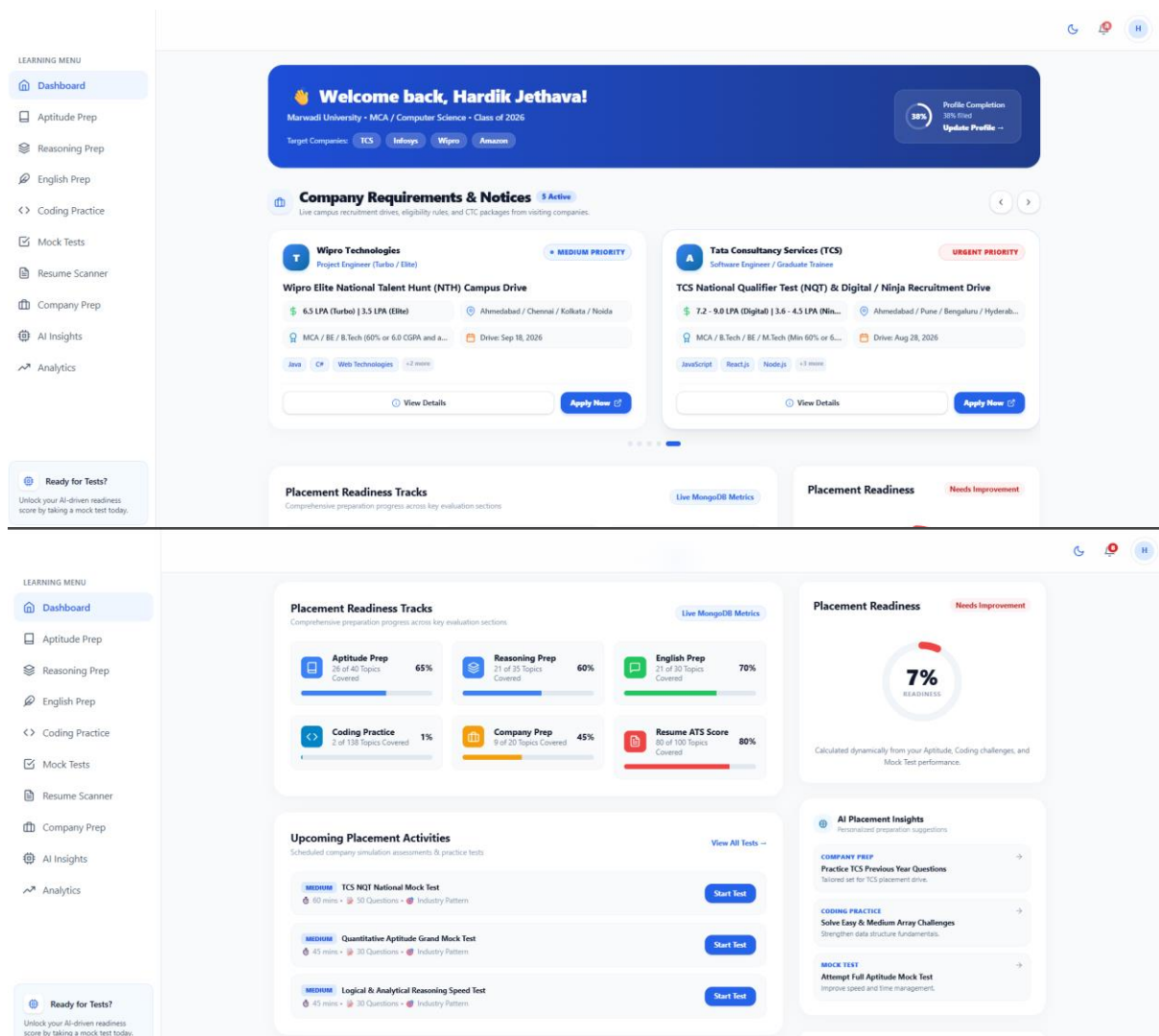


Figure 2.8.3 Student Dashboard Screen

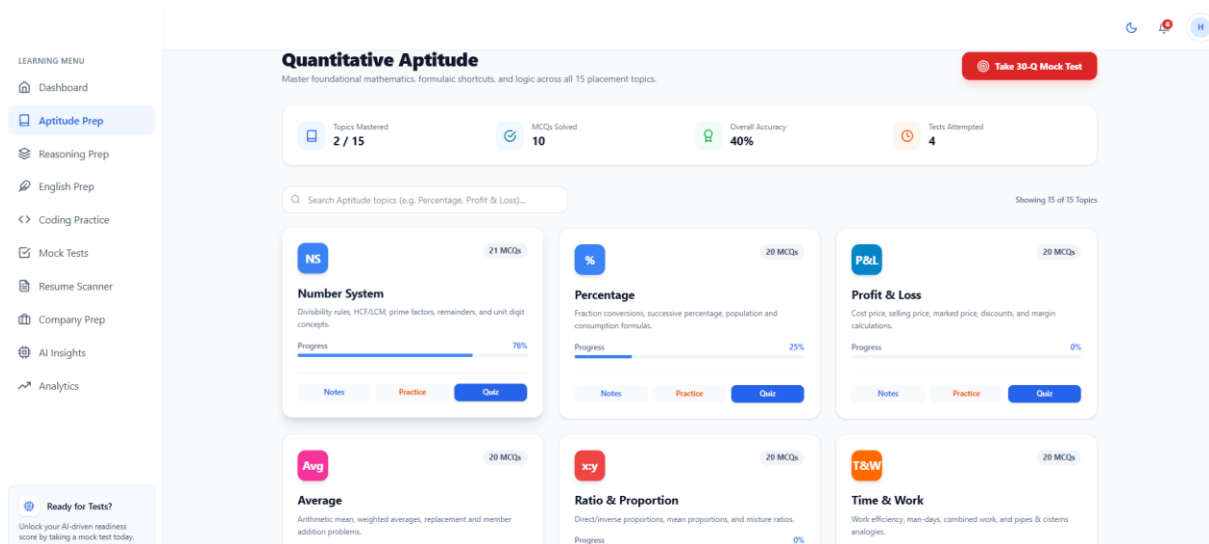


Figure 5.8.4 Aptitude Pre

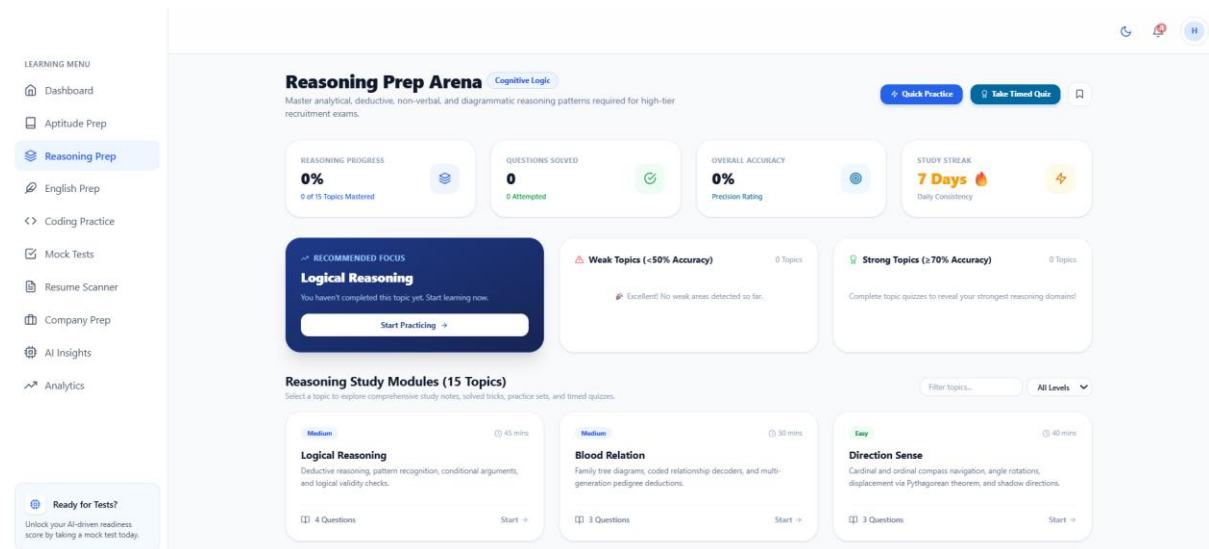


Figure 5.8.5 Reasoning Pre

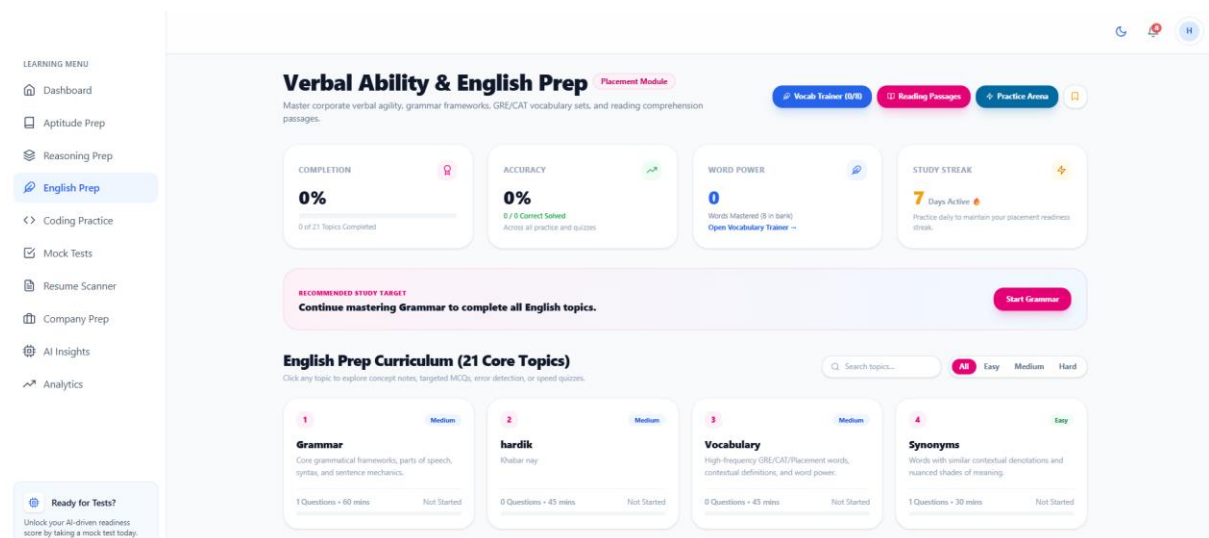


Figure 5.8.6 verbal Ability & English Pre

Figure 5.8.7 Coding Practise

Figure 5.8.8 Mock Test

Figure 5.8.9 Resume Scanning

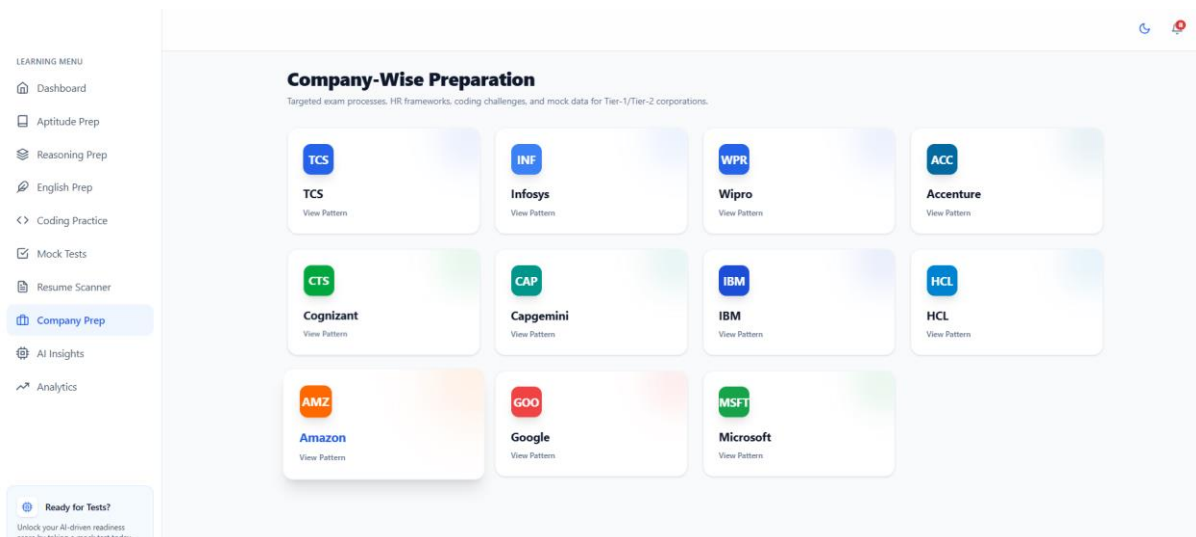


Figure 5.8.10 Company Preparation

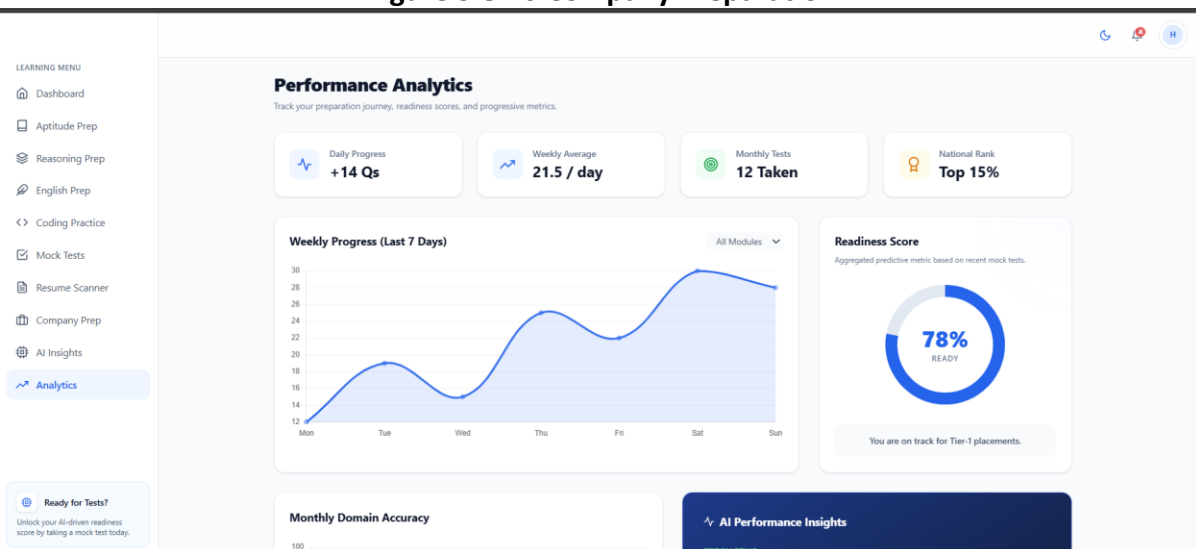


Figure 5.8.11 Performance Analysis

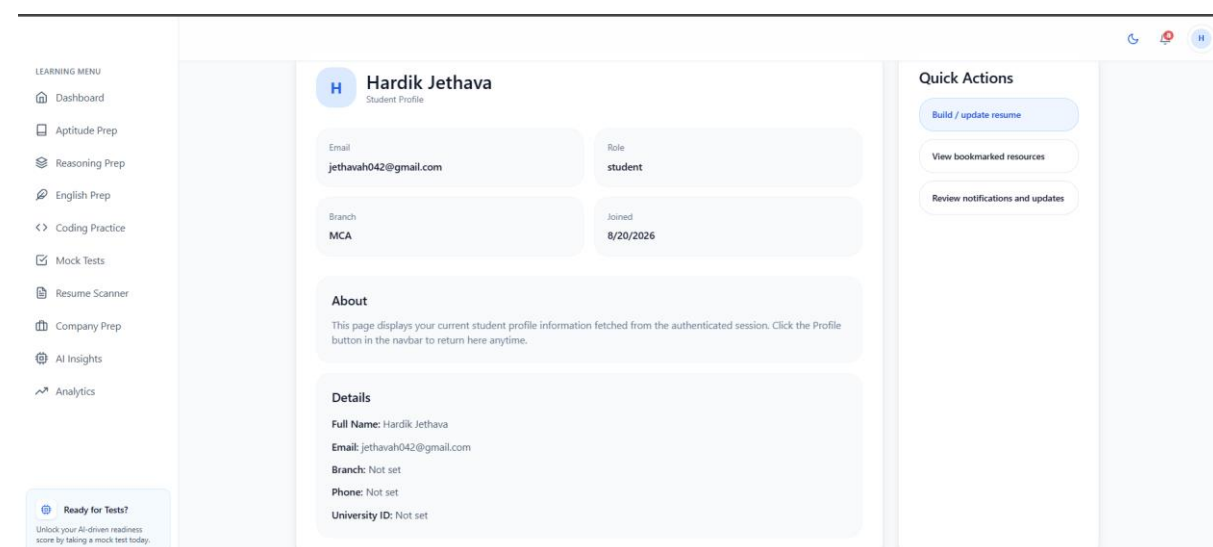
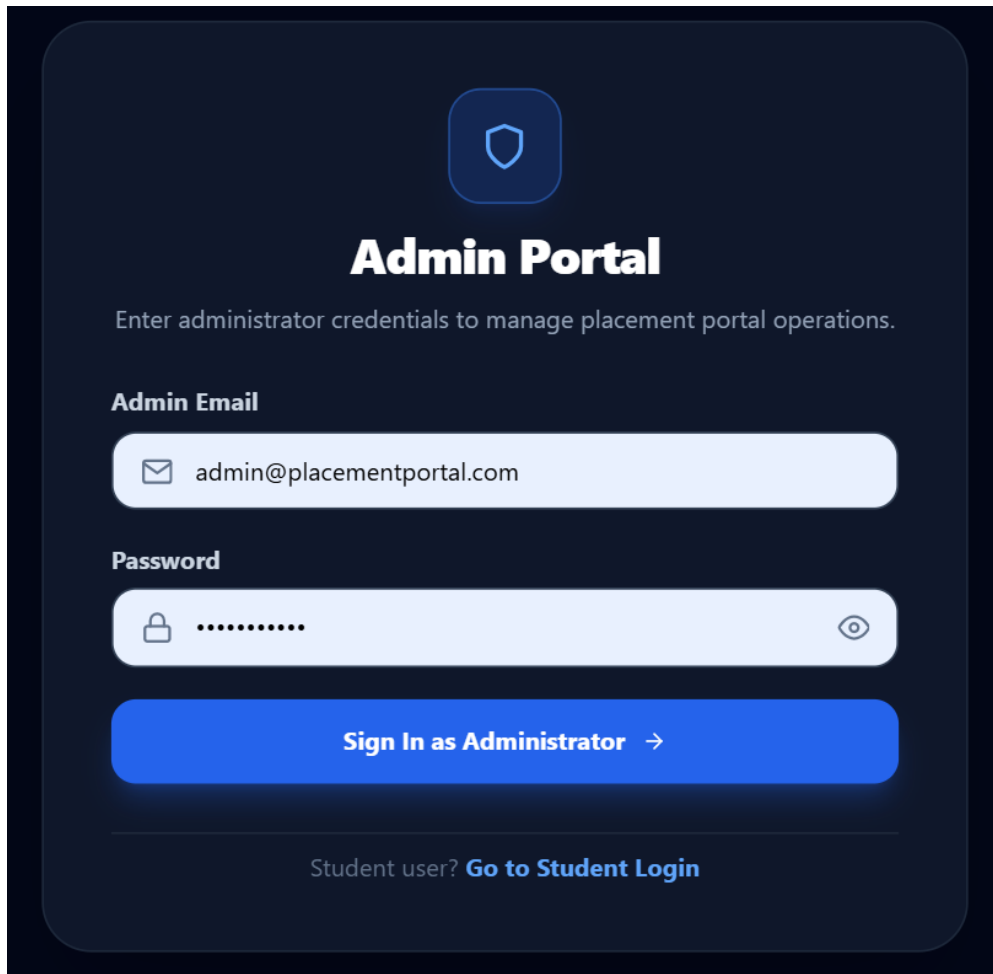


Figure 5.8.12 Student Profiles

5.9 Admin Menu-Screen Design:



The Admin Portal login screen features a dark blue background with a shield icon in a rounded square at the top center. Below the icon, the text "Admin Portal" is displayed in a large, bold, white font. Underneath, a subtitle reads "Enter administrator credentials to manage placement portal operations." The login form consists of two input fields: "Admin Email" with the placeholder "admin@placementportal.com" and "Password" with a masked password ".....". A blue "Sign In as Administrator" button with a right-pointing arrow is positioned below the password field. At the bottom, a link "Student user? Go to Student Login" is provided.

Admin Portal

Enter administrator credentials to manage placement portal operations.

Admin Email

admin@placementportal.com

Password

.....

Sign In as Administrator →

Student user? [Go to Student Login](#)

Figure 5.9.1 Admin Login

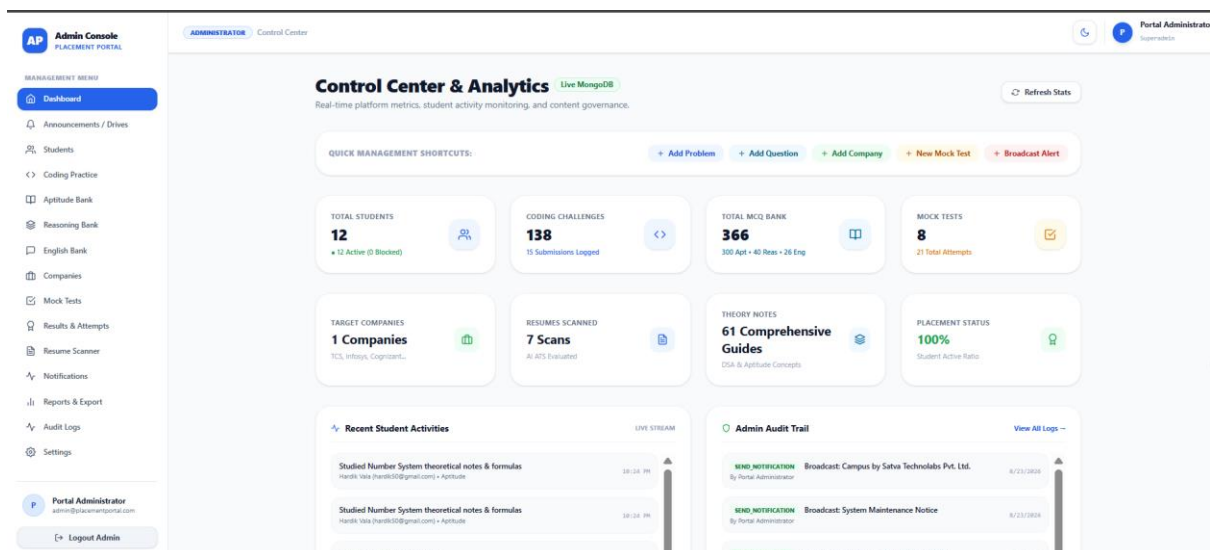


Figure 5.9.2 Admin Dashboard

AP Admin Console

PLACEMENT PORTAL

ADMINISTRATOR Control Center

Portal Administrator

MANAGEMENT MENU

Dashboard

Announcements / Drives

Students

Coding Practice

Aptitude Bank

Reasoning Bank

English Bank

Companies

Mock Tests

Results & Attempts

Resume Scanner

Notifications

Reports & Export

Audit Logs

Settings

Portal Administrator

admin@placementportal.com

Logout Admin

Company Requirements & Announcements

Admin Governance

+ New Announcement

Broadcast upcoming recruitment drives, eligibility rules, and CTC packages to the student dashboard.

All Statuses All Priorities Search company, title, skill

Showing 5 of 5 announcements

Figure 5.9.3 Announcement

AP Admin Console

PLACEMENT PORTAL

ADMINISTRATOR Control Center

Portal Administrator

MANAGEMENT MENU

Dashboard

Announcements / Drives

Students

Coding Practice

Aptitude Bank

Reasoning Bank

English Bank

Companies

Mock Tests

Results & Attempts

Resume Scanner

Notifications

Reports & Export

Audit Logs

Settings

Portal Administrator

admin@placementportal.com

Logout Admin

Student Management

12 Candidates

Monitor registered candidate profiles, verify access status, and manage portal permissions.

All Active Blocked

Search name, email, college...

Figure 5.9.4 Student Management

AP Admin Console

PLACEMENT PORTAL

ADMINISTRATOR Control Center

Portal Administrator

MANAGEMENT MENU

Dashboard

Announcements / Drives

Students

Coding Practice

Aptitude Bank

Reasoning Bank

English Bank

Companies

Mock Tests

Results & Attempts

Resume Scanner

Notifications

Reports & Export

Audit Logs

Settings

Portal Administrator

admin@placementportal.com

Logout Admin

Coding Practice Control

DSA Management

Challenges (136) Topic Notes (11) Interview Qs (122)

Maintain programming challenges, algorithms, starter templates, theory notes, and technical interview questions.

All Topics All Difficulties Search challenges...

+ New Coding Problem

Figure 5.9.5 Coding Practise

AP Admin Console
PLACEMENT PORTAL

MANAGEMENT MENU

Dashboard
Announcements / Drives
Students
Coding Practice
Aptitude Bank
Reasoning Bank
English Bank
Companies
Mock Tests
Results & Attempts
Resume Scanner
Notifications
Reports & Export
Audit Logs
Settings

Portal Administrator
admin@placementportal.com
Logout Admin

ADMINISTRATOR Control Center

Aptitude Question Bank 300 Questions
Manage practice questions, mock test pools, difficulty weights, and step-by-step explanations.

Aptitude Reasoning English

Easy Medium Hard

Search question text...
New Question

Question	Topic	Difficulty	Correct Answer	Marks	Actions
The decimal number 1.23657657657 is equal to the rational number	Number System	Medium	✓ 123534/99900	+1 / -0.25	
A and B invest Rs. 19000 and Rs. 38000 in a venture. At the end of the year, total profit is Rs. 57000. Find A's share.	Partnership	Easy	✓ Rs. 19000	+1 / -0.25	
A and B invest Rs. 20000 and Rs. 40000 in a venture. At the end of the year, total profit is Rs. 60000. Find A's share.	Partnership	Medium	✓ Rs. 20000	+1 / -0.25	
A and B invest Rs. 17000 and Rs. 34000 in a venture. At the end of the year, total profit is Rs. 51000. Find A's share.	Partnership	Easy	✓ Rs. 17000	+1 / -0.25	
A and B invest Rs. 11000 and Rs. 26000 in a venture. At the end of the year, total profit is Rs. 39000. Find A's share.	Partnership	Easy	✓ Rs. 13000	+1 / -0.25	
A and B invest Rs. 16000 and Rs. 32000 in a venture. At the end of the year, total profit is Rs. 48000. Find A's share.	Partnership	Medium	✓ Rs. 16000	+1 / -0.25	
A and B invest Rs. 18000 and Rs. 36000 in a venture. At the end of the year, total profit is Rs. 54000. Find A's share.	Partnership	Hard	✓ Rs. 18000	+1 / -0.25	
A and B invest Rs. 15000 and Rs. 30000 in a venture. At the end of the year, total profit is Rs. 45000. Find A's share.	Partnership	Hard	✓ Rs. 15000	+1 / -0.25	
A and B invest Rs. 14000 and Rs. 28000 in a venture. At the end of the year, total profit is Rs. 42000. Find A's share.	Partnership	Medium	✓ Rs. 14000	+1 / -0.25	

Figure 5.9.6 Aptitude Bank

AP Admin Console
PLACEMENT PORTAL

MANAGEMENT MENU

Dashboard
Announcements / Drives
Students
Coding Practice
Aptitude Bank
Reasoning Bank
English Bank
Companies
Mock Tests
Results & Attempts
Resume Scanner
Notifications
Reports & Export
Audit Logs
Settings

Portal Administrator
admin@placementportal.com
Logout Admin

ADMINISTRATOR Control Center

Reasoning Prep Governance Admin Module
Analytics & Hub Topics (15) Question Bank (40) Study Guides (15) Reports (4)

TOTAL REASONING TOPICS
15
15 Active

QUESTIONS BANK
40
30 Published + 0 Draft

ACTIVE LEARNERS
2
3 Assessment Logs

AVERAGE ACCURACY
29%
Across All Students

Reasoning Curriculum Matrix (15 Modules)
Add Topic

Topic Name	Difficulty	Questions Count	Study Guide	Status	Actions
Logical Reasoning	Medium	4 Questions	✓ Active Guide	active	
Blood Relation	Medium	3 Questions	✓ Active Guide	active	
Direction Sense	Easy	4 Questions	✓ Active Guide	active	
Coding Decoding	Medium	3 Questions	✓ Active Guide	active	
Seating Arrangement	Hard	3 Questions	✓ Active Guide	active	
Syllogism	Hard	3 Questions	✓ Active Guide	active	
Analogy	Easy	2 Questions	✓ Active Guide	active	
Series	Medium	2 Questions	✓ Active Guide	active	

Figure 5.9.7 Reasoning Bank

AP Admin Console
PLACEMENT PORTAL

MANAGEMENT MENU

Dashboard
Announcements / Drives
Students
Coding Practice
Aptitude Bank
Reasoning Bank
English Bank
Companies
Mock Tests
Results & Attempts
Resume Scanner
Notifications
Reports & Export
Audit Logs
Settings

Portal Administrator
admin@placementportal.com
Logout Admin

ADMINISTRATOR Control Center

English Prep Governance Admin Module
Analytics & Hub Topics (21) Question Bank (26) Vocab (8) RC Passages (2) Guides (20) Reports (4)

TOTAL ENGLISH TOPICS
21
21 Active Modules

QUESTIONS BANK
26
16 Published + 0 Draft

VOCABULARY WORDS
8
2 Reading Passages

AVERAGE ACCURACY
48%
8 Assessment Logs

Verbal Ability Curriculum Matrix (21 Modules)
Add Topic

Topic Name	Difficulty	Questions Count	Study Guide	Status	Actions
Grammar	Medium	1 Questions	✓ Active Guide	active	
Idioms	Medium	0 Questions	No Guide	active	
Vocabulary	Medium	0 Questions	✓ Active Guide	active	
Synonyms	Easy	1 Questions	✓ Active Guide	active	
Antonyms	Easy	1 Questions	✓ Active Guide	active	
Idioms	Medium	1 Questions	✓ Active Guide	active	
One Word Substitution	Easy	2 Questions	✓ Active Guide	active	

Figure 5.9.8 English Bank

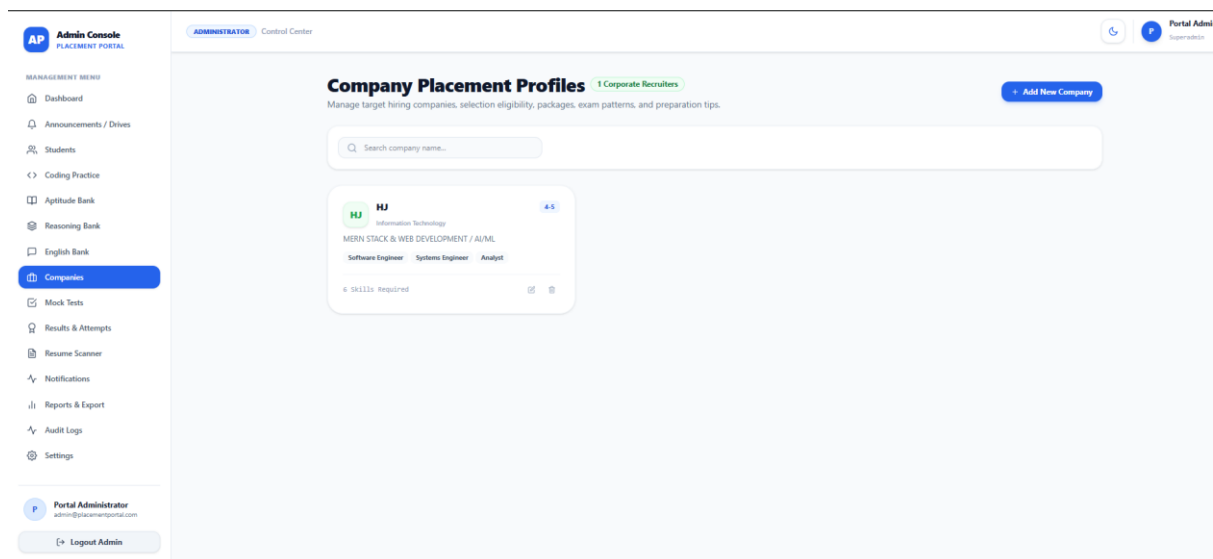


Figure 5.9.9 Companies

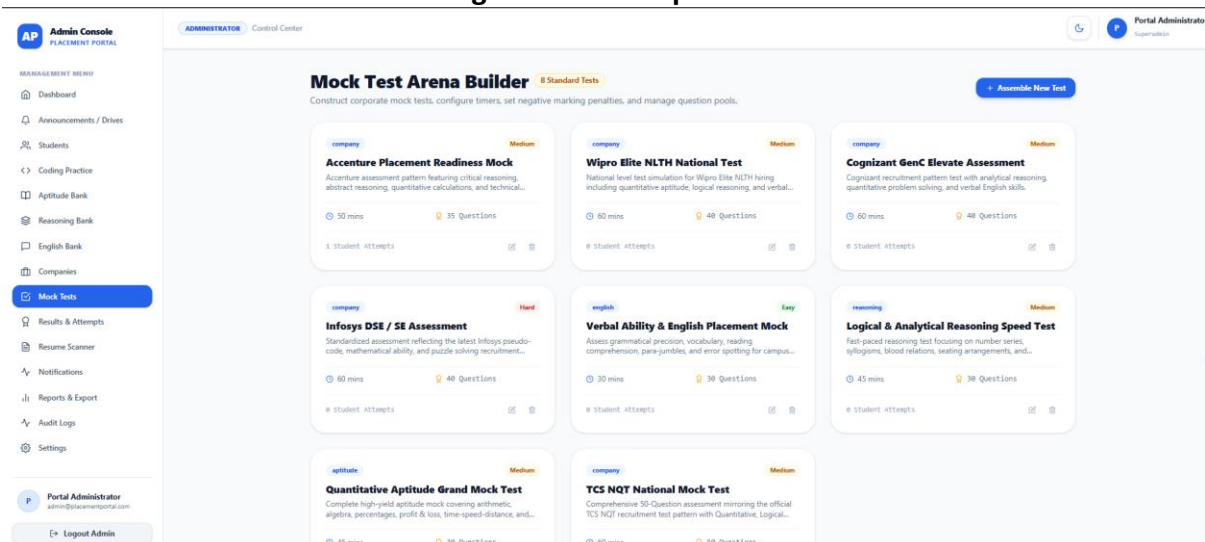


Figure 5.9.10 Mock Test

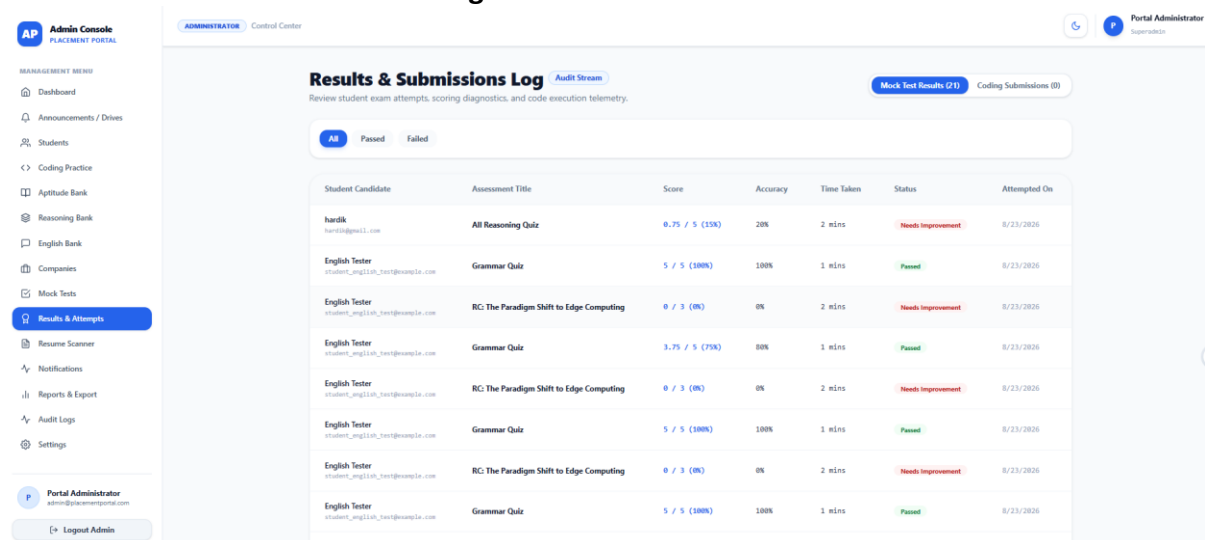


Figure 5.9.11 Results & Attempt

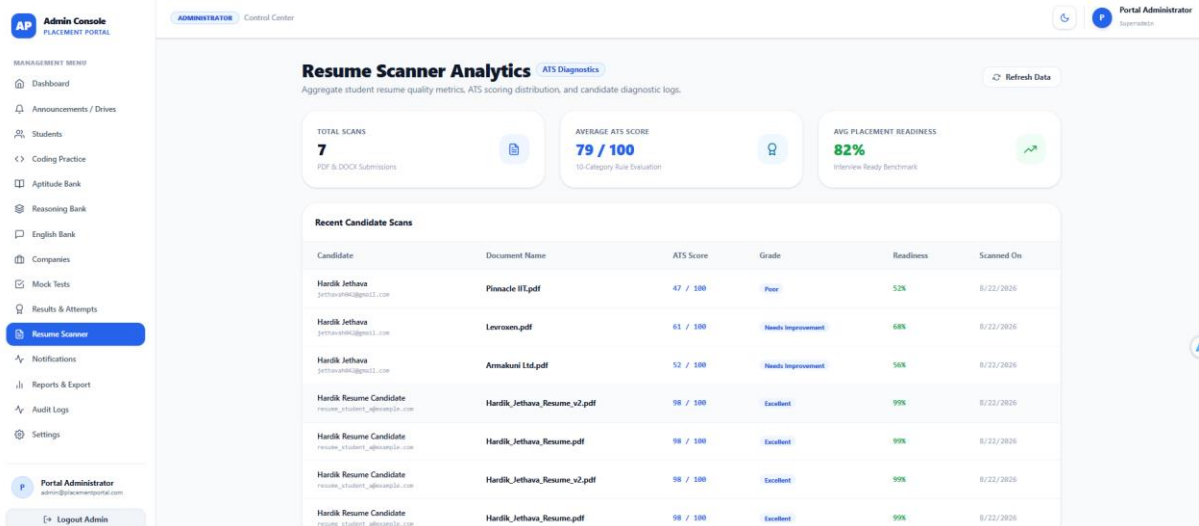


Figure 5.9.12 Resume Scanning

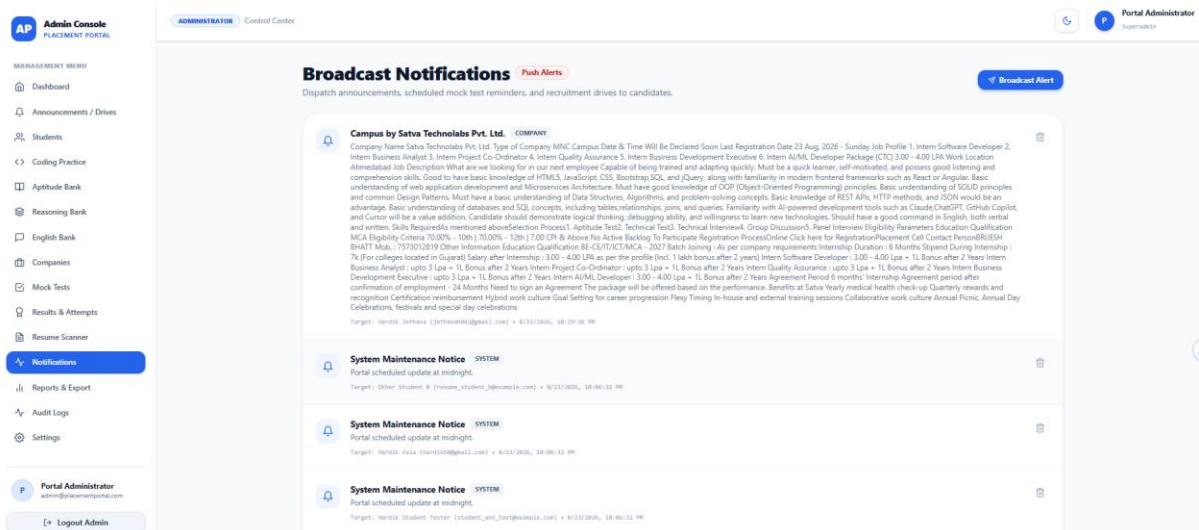


Figure 5.9.13 Notification

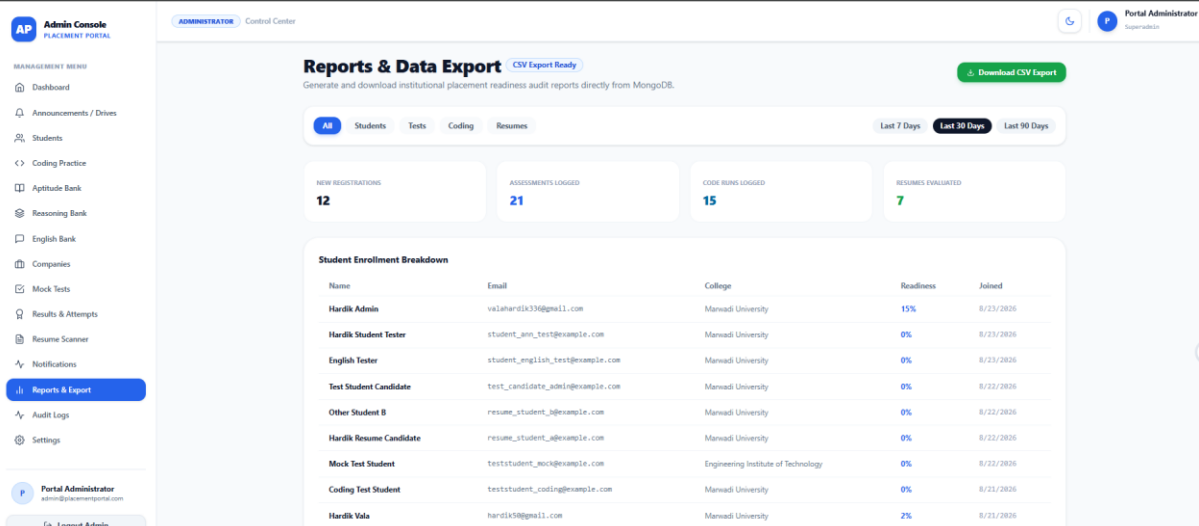


Figure 5.9.14 Report Data & Export

AP

Admin Console

PLACEMENT PORTAL

ADMINISTRATOR Control Center

Portal Administrator

MANAGEMENT MENU

Dashboard

Announcements / Drives

Students

Coding Practice

Aptitude Bank

Reasoning Bank

English Bank

Companies

Mock Tests

Results & Attempts

Resume Scanner

Notifications

Reports & Export

Audit Logs

Settings

Portal Administrator

Logout Admin

Administrator Audit Logs

186 Records

Complete chronological record of all administrative operations, governance interventions, and content revisions.

Timestamp	Admin User	Action Executed	Target Entity	Operation Details
8/23/2025, 10:29:36 PM	Portal Administrator	SEND_NOTIFICATION	Notification	Broadcast: Campus by Satva Technolabs Pvt. Ltd.
8/23/2025, 10:06:32 PM	Portal Administrator	SEND_NOTIFICATION	Notification	Broadcast: System Maintenance Notice
8/23/2025, 10:06:32 PM	Portal Administrator	DELETE MOCK_TEST	Test	Deleted Admin Automated Assessment Test
8/23/2025, 10:06:32 PM	Portal Administrator	CREATE MOCK_TEST	Test	Created mock test Admin Automated Assessment Test
8/23/2025, 10:06:32 PM	Portal Administrator	DELETE_COMPANY	Company	Deleted AlphaCorp Admin Test
8/23/2025, 10:06:32 PM	Portal Administrator	CREATE_COMPANY	Company	Created AlphaCorp Admin Test
8/23/2025, 10:06:32 PM	Portal Administrator	DELETE_QUESTION	Question	Deleted question
8/23/2025, 10:06:32 PM	Portal Administrator	CREATE_QUESTION	Question	Created Aptitude Q in Percentage
8/23/2025, 10:06:32 PM	Portal Administrator	DELETE_CODING_PROBLEM	CodingProblem	Deleted Admin Verification Matrix Challenge
8/23/2025, 10:06:32 PM	Portal Administrator	UPDATE_CODING_PROBLEM	CodingProblem	Updated Admin Verification Matrix Challenge
8/23/2025, 10:06:32 PM	Portal Administrator	CREATE_CODING_PROBLEM	CodingProblem	Created Admin Verification Matrix Challenge
8/23/2025, 10:06:32 PM	Portal Administrator	UNBLOCK_STUDENT	User	Reason: Reinstated
8/23/2025, 10:06:32 PM	Portal Administrator	BLOCK_STUDENT	User	Reason: Assessment policy violation test

Figure 5.9.15 Audit Logs

AP

Admin Console

PLACEMENT PORTAL

ADMINISTRATOR Control Center

Portal Administrator

MANAGEMENT MENU

Dashboard

Announcements / Drives

Students

Coding Practice

Aptitude Bank

Reasoning Bank

English Bank

Companies

Mock Tests

Results & Attempts

Resume Scanner

Notifications

Reports & Export

Audit Logs

Settings

Portal Administrator

Logout Admin

Administrator Settings

Account Security

Manage administrator profile credentials, security configurations, and portal authentication.

Admin Profile Information

Update display name and contact email.

Full Name

Portal Administrator

Email Address

admin@placementportal.com

Save Profile Changes

Security & Password

Change your administrator access passphrase.

Current Password

New Password (8+ chars)

Update Password

Figure 5.9.16 Settings

6. SYSTEM TESTING

Software testing was conducted following IEEE 829 Standard Test Documentation guidelines. Testing was performed to verify the correctness, reliability, usability, and functionality of the Placement Preparation Portal.

Both positive test cases (PASS) and negative test cases (FAIL/Exception Handling) were considered. The testing covered authentication, student modules, preparation modules, mock tests, coding practice, Resume Scanner, Admin Panel, and analytics.

Module 1: User Registration Testing

The User Registration module was tested to ensure that new students can successfully create an account using valid information and that invalid or incomplete information is properly rejected.

Test Case ID	Test Scenario	Test Input	Expected Result	Status
REG-01	Register with valid details	Valid name, email, password	Account should be created successfully	PASS
REG-02	Empty name field	Name left blank	Validation message should be displayed	PASS
REG-03	Empty email field	Email left blank	Validation message should be displayed	PASS
REG-04	Invalid email format	student@	Invalid email message should be displayed	PASS
REG-05	Empty password	Password left blank	Password validation message should appear	PASS
REG-06	Weak password	Short/weak password	System should reject the password	PASS
REG-07	Password mismatch	Different password and confirmation password	Mismatch error should be displayed	PASS
REG-08	Existing email	Already registered email	System should prevent duplicate registration	PASS
REG-09	Valid registration with optional fields	Valid required and optional information	Account should be created successfully	PASS
REG-10	Database connection failure	Registration request when database unavailable	Appropriate error message should be displayed	PASS

Table 6.1 User Registration Testing

Testing Result

The User Registration module successfully validates user input and handles invalid registration scenarios. Duplicate accounts and incorrect input are rejected, while valid student information is stored successfully in the database.

Module 2: User Authentication & Login Testing:

The User Authentication and Login module was tested to ensure that registered students and administrators can securely access the system using valid credentials. Testing also verified that invalid credentials, unauthorized access, and authentication errors are handled appropriately.

Test Case ID	Test Scenario	Test Input	Expected Result	Status
AUTH-01	Login with valid student credentials	Registered email and correct password	Student should be successfully logged in and redirected to Student Dashboard	PASS
AUTH-02	Login with valid admin credentials	Admin email and correct password	Admin should be successfully logged in and redirected to Admin Dashboard	PASS
AUTH-03	Incorrect password	Registered email with wrong password	Login should be rejected with an appropriate error message	PASS
AUTH-04	Unregistered email	Email not present in database	System should display invalid credentials error	PASS
AUTH-05	Empty email field	Email left blank	Email validation message should be displayed	PASS
AUTH-06	Empty password field	Password left blank	Password validation message should be displayed	PASS
AUTH-07	Invalid email format	Incorrect email format	System should display email validation error	PASS
AUTH-08	Empty login form	Both fields left blank	Required field validation messages should be displayed	PASS
AUTH-09	Student accessing Admin Panel	Student token attempts admin route	Access should be denied due to role-based authorization	PASS
AUTH-10	Admin accessing Student features	Admin token attempts restricted student route	Unauthorized access should be prevented where applicable	PASS
AUTH-11	Logout testing	Click Logout	User session/token should be cleared and user redirected to Login page	PASS
AUTH-12	Access protected page without login	Directly open protected URL	User should be redirected to Login page	PASS
AUTH-13	Invalid/expired JWT token	Expired or invalid authentication token	Request should be rejected and user should be asked to authenticate again	PASS

AUTH-14	Login after logout	Valid credentials after logout	User should be able to log in successfully again	PASS
AUTH-15	Database/server unavailable	Login request when backend/database is unavailable	System should display an appropriate error instead of crashing	PASS

Table 6.2 User Authentication & Login Test Case

Testing Result

The User Authentication and Login module successfully authenticates registered users and provides appropriate access according to their roles. Invalid credentials, unauthorized requests, expired authentication tokens, and protected-route access are handled correctly, ensuring secure access to the Placement Preparation Portal.

Module 3: Placement Content & Question Bank Management Testing:

Test Case ID	Test Scenario	Test Input	Expected Result	Status
CNT-01	Add new preparation note	Valid title, subject, topic, and content	Note should be successfully added	PASS
CNT-02	Add question with valid details	Question, options, answer, explanation	Question should be stored successfully	PASS
CNT-03	Add coding problem	Valid problem statement and difficulty	Coding problem should be created successfully	PASS
CNT-04	View content list	Open Notes/Questions section	All available records should be displayed	PASS
CNT-05	Edit existing content	Modify title/content/question	Updated information should be saved successfully	PASS
CNT-06	Delete content	Select an existing record and delete	Selected record should be removed	PASS
CNT-07	Required field validation	Submit form with required fields empty	Validation messages should be displayed	PASS
CNT-08	Invalid question data	Missing options or correct answer	System should reject invalid question data	PASS
CNT-09	Difficulty validation	Select Easy/Medium/Hard	Selected difficulty should be stored correctly	PASS
CNT-10	Search content	Enter valid keyword	Matching notes/questions should be displayed	PASS
CNT-11	Filter content	Select subject/topic/difficulty	Relevant records should be displayed	PASS
CNT-12	Student content access	Student opens preparation module	Admin-created content should be displayed to the student	PASS
CNT-13	Updated content synchronization	Admin edits existing content	Updated content should be reflected in Student Panel	PASS
CNT-14	Unauthorized content management	Student attempts Admin CRUD operation	Access should be denied	PASS
CNT-15	Database/server error	Database unavailable during operation	Appropriate error message should be displayed without crashing	PASS

Table 6.3 Product & Inventory Management Testing

Testing Result

The Placement Content and Question Bank Management module successfully performs CRUD operations for preparation content. The testing confirms that administrators can manage notes, questions, and coding problems, while students can access the updated.

Module 4: Mock Test & Assessment Testing

Test Case ID	Test Scenario	Test Input	Expected Result	Status
TEST-01	Open available mock tests	Student selects Mock Test	Available tests should be displayed	PASS
TEST-02	Start aptitude test	Select Aptitude Mock Test	Test should start successfully	PASS
TEST-03	Start reasoning test	Select Reasoning Mock Test	Test should start successfully	PASS
TEST-04	Start English test	Select English Mock Test	Test should start successfully	PASS
TEST-05	Start coding test	Select Coding Mock Test	Coding test should start successfully	PASS
TEST-06	Start mixed placement test	Select Mixed Test	Mixed test should load correctly	PASS
TEST-07	Verify 30 questions	Start configured 30-question test	Exactly 30 questions should be displayed	PASS
TEST-08	Question navigation	Move between questions	Previous/Next navigation should work correctly	PASS
TEST-09	Select answer	Select an option	Selected answer should be saved	PASS
TEST-10	Change answer	Select another option	Previous answer should be replaced correctly	PASS
TEST-11	Skip question	Move without selecting answer	Question should remain unanswered	PASS
TEST-12	Timer functionality	Start timed test	Countdown timer should start correctly	PASS
TEST-13	Timer expiration	Allow timer to reach zero	Test should be automatically submitted	PASS
TEST-14	Manual submission	Click Submit Test	Test should be submitted successfully	PASS
TEST-15	Correct answer scoring	Submit correct answers	Correct answers should receive marks	PASS
TEST-16	Incorrect answer scoring	Submit incorrect answers	Incorrect answers should be evaluated correctly	PASS
TEST-17	Negative marking	Submit incorrect answer in negative-marking test	Applicable negative marks should be deducted	PASS
TEST-18	Result generation	Complete test	Score, correct, incorrect, and skipped answers should be displayed	PASS
TEST-19	Result persistence	Complete and submit test	Result should be saved in MongoDB	PASS
TEST-20	Performance analysis	View completed test result	Accuracy, percentage, and performance analysis should be displayed	PASS
TEST-	Test history	Open previous test	Previous test results	PASS

21		attempts	should be displayed	
TEST-22	Unauthorized test submission	Invalid/expired authentication token	Submission should be rejected	PASS
TEST-23	Refresh during test	Refresh test page	System should handle the session appropriately without corrupting the test result	PASS
TEST-24	Server/database failure	Submit test when server/database unavailable	Appropriate error should be displayed	PASS

Table 6.4 Mock Test & Assessment Testing

Testing Result

The Mock Test and Assessment module successfully performs test initialization, question navigation, timer management, answer submission, scoring, negative marking, automatic submission, and result generation. Test results are stored in the database and are used to provide students with detailed performance analysis and progress tracking.

7. CONCLUSION

The **Placement Preparation Portal** was developed as a comprehensive web-based platform to help students prepare effectively for campus placements through a single centralized system. The project integrates multiple placement preparation activities, including **Aptitude, Reasoning, English, Programming Languages, Coding Practice, Mock Tests, Company Preparation, Resume Scanner, AI-assisted Interview Preparation, and Performance Analytics**.

The system provides students with topic-wise study material, practice questions, quizzes, coding challenges, and timed mock tests. Students can view their test results, track their preparation progress, identify weak areas, and improve their overall placement readiness. The Resume Scanner further helps students review their resumes and receive suggestions for improvement.

The **Admin Panel** provides centralized and dynamic management of students, notes, questions, coding problems, companies, mock tests, and results. This allows preparation content to be updated without modifying the application manually.

The project was developed using the **MERN Stack**, consisting of React.js for the frontend, Node.js and Express.js for backend services, and MongoDB with Mongoose for data management. **JWT authentication** and role-based access control provide secure access for students and administrators.

Overall, the Placement Preparation Portal provides a structured, accessible, and interactive environment for student placement preparation. It reduces the need to use multiple platforms and brings learning, practice, assessment, resume improvement, and performance tracking together in one system. The project can also be further enhanced in the future with advanced AI interview simulations, additional company question banks, real-time coding evaluation, personalized learning recommendations, and mobile application support.

8. LEARNING DURING PROJECT WORK

During the development of the **Placement Preparation Portal**, several technical and practical skills were gained through the design, development, testing, and integration of the system.

1. **MERN Stack Development**
Gained practical experience in developing a complete web application using React.js, Node.js, Express.js, MongoDB, and Mongoose.
2. **Frontend Development**
Learned how to create responsive and user-friendly interfaces using React.js, reusable components, routing, forms, dashboards, and modern UI techniques.
3. **Backend Development**
Developed an understanding of creating REST APIs using Node.js and Express.js and connecting different frontend modules with backend services.
4. **Database Management**
Learned how to design MongoDB collections and manage student profiles, questions, notes, tests, results, coding problems, companies, resumes, and progress data.
5. **Authentication and Authorization**
Gained practical knowledge of JWT-based authentication, password security using bcryptjs, protected routes, and role-based access for Students and Admins.
6. **API Integration**
Learned how to connect frontend components with backend APIs using Axios and handle data requests, responses, validation, and errors.
7. **Dynamic Content Management**
Learned how an Admin Panel can dynamically create, update, delete, and manage preparation content that is displayed to students.
8. **Testing and Debugging**
Developed skills in identifying errors, testing positive and negative scenarios, debugging frontend and backend issues, and improving system reliability.
9. **Mock Test and Result Processing**
Learned how to implement timed tests, question navigation, answer submission, scoring, negative marking, result generation, and performance analysis.
10. **Project Architecture**
Gained practical understanding of MVC architecture, modular development, reusable components, API organization, and maintaining a structured project.
11. **Problem-Solving Skills**
The project helped improve logical thinking and problem-solving skills by handling real-world requirements, database relationships, authentication issues, and module integration.
12. **Project Management**
Learned to divide a large application into smaller modules and develop, test, and integrate each module systematically.
13. **Documentation Skills**
Gained experience in preparing technical documentation, system design diagrams, testing documentation, and project reports.
14. **Deployment and Production Concepts**
Developed an understanding of environment variables, production configuration, API security, database connectivity, and deployment requirements.

8.1 Future Enhancement:

The Placement Preparation Portal can be further enhanced with advanced features to make the system more intelligent, personalized, and useful for students.

1. **Advanced AI Interview System**
An AI-based virtual interviewer can conduct real-time technical and HR interviews and provide feedback on student performance.
2. **Personalized Learning Recommendations**
The system can analyze a student's test results and automatically recommend topics, questions, and study material based on their weak areas.
3. **AI-Based Resume Analysis**
The Resume Scanner can be enhanced with advanced ATS analysis, keyword matching, resume scoring, and job-specific resume recommendations.
4. **Online Coding Compiler**
A real-time coding environment can be integrated so students can write, compile, and execute programs directly within the portal.
5. **More Company-Specific Content**
The company preparation module can be expanded with regularly updated aptitude, coding, technical, and HR questions from more companies.
6. **Mobile Application**
A dedicated Android/iOS application can be developed to allow students to access preparation material and tests from mobile devices.
7. **Real-Time Leaderboard**
A real-time leaderboard can be introduced to encourage healthy competition among students based on test scores, coding performance, and preparation progress.
8. **Advanced Analytics**
More detailed analytics can be added to identify performance trends, topic-wise weaknesses, study patterns, and overall placement readiness.
9. **Notifications and Reminders**
The system can provide personalized notifications for upcoming tests, recommended practice, study reminders, and placement-related updates.
10. **Multilingual Support**
The portal can support multiple languages so that students can access preparation content in their preferred language.
11. **Cloud-Based Scalability**
The application can be deployed using cloud infrastructure to support a larger number of students and provide reliable access.

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